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# ESSAYS ON MARKET VOLATILITY

ROGERIO BATISTA ADELINO

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# ESSAYS ON MARKET VOLATILITY

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ROGERIO BATISTA ADELINO

# ESSAYS ON MARKET VOLATILITY

Tese apresentada ao Programa de Pós-Graduação em Administração de Empresas da Universidade Presbiteriana Mackenzie, como requisito parcial à obtenção de título de Doutor em Administração de Empresas.

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# RESUMO

O entendimento do tema de volatilidade e sua dinâmica é crucial para a teoria e pratica do mercado financeiro. Neste campo, no qual a variável aleatória considerada é geralmente o retorno de um ativo, a volatilidade pode ser vista como a dispersão em relação a uma medida de centralidade. Essa ideia tem impacto direto tanto para a academia como para a pratica, desde o simples cálculo do retorno composto, precificação de ativos e gerenciamento de portifólios. Ademais, a volatilidade também deve ser vista como um processo dinâmico, sujeito a mudanças ao longo do tempo de acordo com eventos específicos ou tendencias, o que afeta, por exemplo, a capacidade de proteção oferecida pela diversificação de ativos.

Recentemente, a capacidade de processamento e a disponibilidade de informação abriram uma nova perspectiva de estudo. O uso de dados de alta-frequência habilitou um campo econométrico novo. Na perspectiva tradicional, modelos econométricos são empregados para entender elementos da série, tais como tendencia ou estocasticidade. Com o desenvolvimento para altas-frequências, no computo de segundos ou até mesmo milissegundos, elementos cruciais distintos emergem na forma de rupturas na série de tempo (pulos).

Essa pesquisa visa contribuir para o estoque teórico atual de volatilidade e risco por avançar no entendimento da interconexão entre medidas de volatilidade tradicionais e de alta-frequência, evoluir na identificação de pulos e seus condicionantes, desenvolver ligação entre eficiência e risco microestrutural, prover exemples inovadores do emprego dos models e aproximar a teoria da pratica da indústria de investimentos.

Para a obtenção destes objetivos, o trabalho inclui três artigos desenhados para prover novidades no cálculo e uso de volatilidade na pratica de investimento, apontando para a amplitude deste tema. O primeiro artigo foca na capacidade de hedging de índices de volatilidade em portifólios de ações. O segundo artigo utiliza dados de alta-frequência para demonstrar diferenças de risco entre grandes e pequenas empresas abertas. O último artigo avalia o mercado acionário brasileiro em uma perspectiva de ultra-alta-frequência.

A pesquisa demonstra implicações na capacidade de proteção de instrumentos de volatilidade, aponta para a presença de pulos em alta-frequência e delineia condições de eficiência nas quais pulos ou ruído são prevalentes, com consequências em termos de eficiência de mercado e heterogeneidade de risco.

# ABSTRACT

The understanding of volatility and its dynamics plays a crucial role in financial markets and theory. In the field of finance and investments where the general random variable being considered is the return of an asset, volatility can be understood as the dispersion of the returns in relation to a centrality measure. This idea has direct implications for both the academy and the practice, from a simple compounding return appraisal to the pricing of financial assets and portfolio management. Additionally, volatility also can be analyzed as a dynamic process, with a changing nature over time, according to specific events or trends in the market, affecting as an example the protection capacity that diversification is able to provide.

In recent years, computing capabilities and information availability opened a new perspective for volatility studies. The computation of data on the high-frequency scale enabled a potentially new econometric framework. In common sampling frequency, such as daily calculations, for example, traditional econometric models can be used to understand the elements of the series, as in the case of drift or randomness. As the model evolves based on the usage of high-frequency data, in the sort of seconds or even shorter frequencies, some other crucial elements in the dynamic of prices in the form of ruptures in the time series (jumps) may be uncovered.

This research project aims to contribute to the existing theoretical framework of volatility and risk by advancing the understanding of the interconnectedness between the traditional and high-frequency volatility measures, evolving the use of jump identification measures and their economic enablers, developing the understanding of market efficiency and microstructure risk, providing innovative examples of real-life modeling, and approximation of the theory with practical strategies in the investment industry.

To achieve these goals, the work includes three articles designed to highlight novelties in the calculation and usage of volatility measures in investment practice, pointing to the broadness of the theme of the study. The first article focuses on the hedging capability of volatility indexes in equity portfolios, the second article utilizes the sort of upper-end of high-frequency data to demonstrate the riskiness differences between large and small-cap equities, and the last article goes beyond in evaluating the Brazilian stock exchange under an ultra-high-frequency framework.

The research depicts implications to hedging using volatility instruments, points to the presence of jumps in high-frequency, and demonstrate efficient conditions in which jumps or noise are frequent in terms of market efficiency and heterogeneity of microstructure riskiness.

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# List of Abbreviations and Acronyms

| | |
|---|---|
| **AR** | **Autoregressive Model** |
| **ARCH** | **Autoregressive Conditional Heteroskedasticity Model** |
| **ARMA** | **Autoregressive and Moving Average Model** |
| **BOVESPA** | **Bolsa de Valores de Sao Paulo (Sao Paulo Stock Exchange)** |
| **CAPM** | **Capital Asset Pricing Model** |
| **CBOE** | **Chicago Board Options Exchange** |
| **CCSA** | **Centro de Ciencias Sociais Aplicadas (Center for Applied Social Sciences)** |
| **CFA** | **Chartered Financial Analyst** |
| **DJIA** | **Dow Jones Industrial Average** |
| **ETF** | **Exchange Traded Funds** |
| **ETN** | **Exchange Traded Notes** |
| **FOMC** | **Federal Open Market Commitee** |
| **FV** | **Future Value** |
| **GARCH** | **Generalized Autoregressive Conditional Heteroskedasticity Model** |
| **IBOVESPA** | **Index of Sao Paulo Stock Exchange** |
| **MA** | **Moving Average Model** |
| **NYSE** | **New York Stock Exchange** |
| **OLS** | **Ordinary Least Squares** |
| **PV** | **Present Value** |
| **S&P 500** | **Index of 500 stocks, published by Standard & Poor's** |
| **VIX** | **CBOE S&P 500 Volatility Index** |
| **VXO** | **CBOE S&P 100 Volatility Index (discontinued)** |

# 1. INTRODUCTION AND RESEARCH PROBLEM

The evaluation of market volatility is of great interest in finance. Formally stated, the volatility can be thought as the dispersion of a certain random variable in relation to its centrality measure. In the field of finance, and more specifically investments, the general random variable being considered is the return of an asset, and volatility is measured as the standard deviation of the returns. The understanding of volatility implications on investing is crucial for both the academy and the practice. Initially, it has a direct impact on the compounding return of an asset, as higher volatility reduces the final compounding returns. In addition, one of the cornerstones in finance states that the expected return on an asset is directly proportional to the risk offered by that asset, meaning that a proper volatility appraisal may be crucial for asset pricing. With effect, pricing methodologies for assets, as it is the example of the Black – Scholes – Merton formula for options, include formally a volatility parameter as an input.

In terms of investment and portfolio management, again volatility becomes an important element of consideration. From a static standpoint, volatility plays a crucial role in portfolio design. As an example, Modern Portfolio Theory combines assets in a diversified portfolio with the goal to minimize risk. In such a framework, diversification can be understood as exposure to different volatility patterns towards a specific risk factor, leveraging the importance of volatility understanding. Additionally, volatility also can be analyzed as a dynamic process, with a changing nature over time, according to specific events or trends in the market. DeFusco et al. (2007), as an example, describe a spike in market volatility related to derivatives expiration days. In another example, practitioners are often concerned about Central Bank interest rate decisions in their activities. Gardner, Scotti & Vega (2022) depict the impact in equity markets as a result of FOMC announcements. Blinder et al. (2022) go further in evaluating the conditions in which central bank communication impacts the general public.

In many aspects, the understanding of market dynamics, including returns and risks (i.e. volatility), represents a central piece of action with academic and professional impact. Despite its huge attention, the study of volatility as an independent subject has boundaries that cannot be easily traced as it may serve a multiplicity of objectives, but also play a crucial role even in cases where volatility is not the theme of study. In such cases, one can easily perceive the amount of theorization produced in finance with objectives such as portfolio design and risk management, for example, including seminal works that transformed capital markets, such as the above-mentioned Markowitz's Modern Portfolio Theory. Moreover, these theories have evolved from the academic landscape into the daily work of portfolio managers, traders, investors, and regulators, in the pursuit of the best investment return–risk profile possible. As a result, an analysis of publications related to volatility in the Financial Analysts Journal, issued by the CFA Institute, a leading professional organization in the field of investments, depicts the importance of volatility in fields like Investment Management Strategies, Portfolio Optimization and Rebalance, Return Analysis and Performance, Active Management, Financial Markets, Bonds, Investment Products and Asset Classes, and Risk Analysis and Management, among others (CFA, 2022).

Several examples can also be raised in the context of evaluating volatility as an independent field within finance and investments. Maybe one of the most important is the CAPM theory, which development started with the seminal works of Sharpe (1964), Lintner (1965), and Mossin (1966), in which the total volatility is decoupled into a market factor and idiosyncratic volatility, with important consequences to diversification.

Loosely put, volatility can be understood as the potential variation of one outcome, in investment often related to returns. As such, the assumptions and calculations, as well as the applicability and limitation of the concept, will also present remarkable diversity. From the descriptive statistics point of view, volatility measurement can assume many forms (range, variance, standard deviation, among others) where the central idea is that it is a dispersion around the mean return in an ex-post sample of returns. On a different angle, volatility can be seen as an ex-ante expectation embedded in the price of assets, as initially depicted in the works of Black & Scholes (1973), and Merton (1973). This approach evolved to the elaboration of volatility indices that are used every day by investors and managers, such as the VIX, issued by the CBOE. New developments in computation capacity and mathematical algorithms have allowed for the appraisal of volatility on an instant-like result, generated by the specific movement of the time series to be able to deal with this challenge in a high-frequency environment, mathematics is still being developed to measure the impact of distinctive features not perceived or solved by traditional econometrics of time. Despite the strong mathematical content, this theme needs to be centralized in finance, due to the fact that the theoretical landscape in which the volatility can be properly understood resides in the field as well as the practical implications in resource allocation and performance management of investments. Over time, several models were created to deal with the measurement of volatility but the problem remains: How can we optimize portfolio returns considering the risk? How can the riskiness of the market be properly measured and forecasted? What is the implication of the changing profile of risk in terms of investment performance? As a result, despite dealing with a mathematical challenge, any research in this field should always keep its ground on the financial theory; but not only, also the financial practice.

A first and logical example can be related to trading. Figure 1.1 depicts a candle bar example of a price movement over a period that is subdivided into six parts and exemplifies the informational differences in the price trends. In the figure, the total period n is divided into 6 transactions that occurred in a 1/6 fraction of period n, therefore resulting in a shorter-term periodicity n/6. The resulting candlestick, formed from the aggregation from individual short-term transactions, simplify the information content. Short-term trading strategies, such as scalping trade, rely already on reaction to very short-term price movements, frequently driven by automatized ordering and colocation in physical points of presence of exchanges.

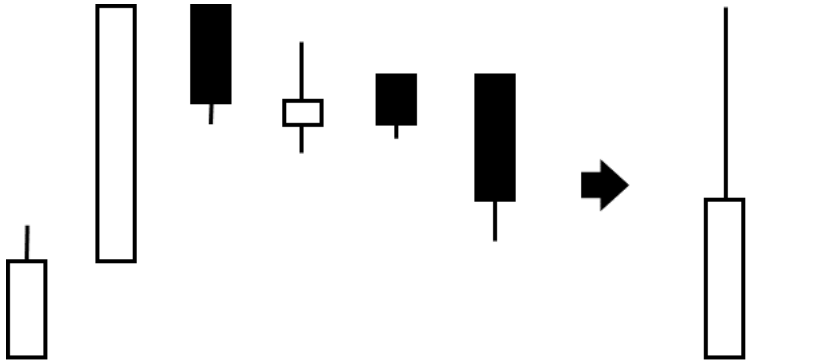
Secondly, the pricing of derivatives is generally calculated with volatility as a central input. The seminal Black, Scholes (1973), and Merton (1973), as an example, design a model for contingent claims where the price path of the underlying security follows a martingale assumption with a normality probability distribution. The volatility of the asset is, then, an input of the broadness of the distribution and is evaluated under the traditional, low-frequency, framework. With the incorporation of short-term price dynamics, an evolved landscape may imply more advanced derivatives pricing models.

In addition to this, investors and professional portfolio managers are constantly seeking for risk control within their portfolios. Since the works of Markowitz (1952), Treynor & Black (1973), and Black & Litterman (1992), portfolios have been optimized considering a measure of average returns, volatility, and correlations (depending on the case, raw measures or active) seeking to control the risk based on diversification while maximizing the expected returns. Again, as the high-frequency approach uncovers new price and volatility dynamics, the very trend may become a relevant input in portfolio design and risk control.

In close relation to the portfolio discussion, hedging strategies also may benefit from an enhanced understanding of price dynamics and their consequent volatility. As the elements within a time series of prices are clear, hedging strategies may become more adherent to the underlying risk of each component. While drift can be more simply hedged, for example, short-term tail movements could be a concern in time series with strong jump activity.

*Figure 1.1*

**Example of Difference in Informational Content of Price Dynamics at Different Frequencies**




From an econometric point of view, when evaluated from a univariate time-series perspective, recent computing capabilities allowed for the computation of data in the high-frequency scale, opening a potentially new field for understanding price dynamics. In common sampling frequency, such as daily calculations, for example, traditional econometric models can be used to understand the elements of the series, as in the case of drift or randomness. As the model evolves based on the usage of high-frequency data, in the sort of seconds or even shorter frequencies, some other crucial elements in the dynamic of prices in the form of ruptures in the time series may be uncovered.

This research project aims to contribute to the existing theoretical framework of volatility and risk. Initially, the work advances the understanding of the interconnectedness between the traditional volatility measures, based on centrality and dispersion econometrics, and the microstructure riskiness of the time series in high-frequency. At this point, the econometric framework for high-frequency is still under development, as well as the understanding of the microstructure risk implication to the traditional volatility framework. This research contributes to the construction of this bridge between those two perspectives. On a consequential level, the articles show a framework to employ the mathematics of jump identification to understand different market structures, such as a factor or specific features related to a group of stocks, pointing to heterogeneity in microstructure risks among assets.

A third contribution is related to the frontier expansion related to jump identification on an ex-post basis and the evaluation of potential implications in the high-frequency time-series itself but also in the efficiency appraisal of specific market elements with consequences in understanding low-frequency

resulting volatility. In such a way, the work provides elements to conclude for heterogeneous microstructure riskiness. The research will also provide innovative examples in which the high-frequency can be explored, in terms of market structure and also by analyzing an emerging market equity index that will contribute to a broader understanding of this theory. At last, the final objective is related to the approximation of the theory with actionable strategies in the investment industry.

To achieve these goals, the work includes three articles designed to highlight novelties in the calculation and usage of volatility measures in investment practice, pointing to the broadness of the theme of the study. In such a way, the research contains three different views of volatility. The first article focuses on the hedging capability of volatility indexes in equity portfolios. As a result, it is based on an ex-ante volatility measurement in a traditional econometrics environment, evaluating the practical use of instruments for hedging equity portfolios. The second article utilizes the sort of upper-end of high-frequency data to demonstrate the riskiness differences between large and small-cap equities. The data utilized is a one-minute time series for the Russell 1000, 2000, and 3000 indices, intending to depict differences in the probability of jump occurrence in the time series. The last article goes beyond in evaluating the Brazilian stock exchange under an ultra-high-frequency framework, where data is collected on 1/100 of a second, aiming at the conceptualization of the riskiness of the market and pointing to enablers of microstructure risk and rupture of the time series. As part of a deductive approach, the research builds up from different models and approaches, categorized by their component elements. This categorization doesn't form a timeline itself, meaning that there's no necessary precedence on some models compared to others, although there's a logical relationship that more sophisticated models are building blocks constructed on top of models with fewer features.

The main body of work, applicable to high-frequency statistics, will be a test of the time-series components presented by Ait-Sahalia & Jacod (2012), which draws a parallel of original applications in physics with the returns of stock markets. The authors utilize light spectrum analysis, a theoretical landscape applicable to the study of the chemical elements in a star, to develop a high-frequency analytical framework for financial markets. In the first application, after filtering for astronomical interferences that cause blurring images, the evaluation of nuclear reactions is the catalyst for understanding the chemical components in the celestial body. In the case of financial markets, market returns can be understood as the result of semi-martingales (nuclear reactions) that may, in reality, be decoupled in each component of the martingale (chemical elements) such as a drift, a 'Brownian Motion', and discontinued jumps.

Part of this research is viable due to an agreement of Mackenzie University with BOVESPA that involves colocation and the provision of real-time data about transactions that occurred within the stock exchange, including closed prices, bid-ask spreads, and volumes.

As such, the research problem proposed is: How to manage volatility, and which are the components of volatility, with specific implications to portfolio risk management arising from this understanding?

To solve this question, some constructs will be operationalized. Initially, the discussion about volatility indices and the potential hedging opportunities directly related to the financial instruments on equity portfolios. In addition, the formation of the time series corresponding to the IBOVESPA and Russell indexes and the prices of components are another cornerstone of the research. In this case, the calculations will thrive to segregate three potential time series components into another set of crucial constructs. These are related to the Drift, the Brownian Motion, and the discontinued Jumps.

Two important features in equity market structures are related to liquidity and breadth of the market. The DJIA is the most traditional equity index in the world and is located in the most liquid stock market in the world, the New York Stock Exchange NYSE. In terms of breadth, despite the huge number of listed companies in the US market, the DJIA is composed of only 30 securities (S&P, 2021). The Sao Paulo Stock Exchange, on the other hand, is a less liquid market and IBOVESPA is composed of 91 companies (B3, 2021). Another important difference resides in the weighting mechanism for both indexes, as the DJIA is price-weighted while IBOVESPA is Market-cap-weighted. These differences are expected to exacerbate the jump component, as the lower liquidity in a broader index (in terms of ticks) should imply more discontinuation of the series, and therefore exacerbate the jump element. For this evaluation, the results yielded by this research may be critically compared to the one achieved by Ait-Sahalia & Jacod (2012).

## 1.1. RESEARCH STRUCTURE

This session will summarize the articles included in this research. Each article will form a specific chapter in this document. Article 1 will be depicted in chapter 2, following this introduction (chapter 1), and so on. The three articles included in this research analyze market volatility from different perspectives. As stated, the main proposition in this effort is to expand the current scientific understanding of the topic respecting different approaches and uses that it might assume, and foremost assure the link between finance theory and investment practice. In this way, instead of prescribing one unique formula to treat the subject, this work thrives to expand knowledge in a way that may be of interest to many parts of the industry, at least partially.

### 1.1.1. ARTICLE 1: TRADITIONAL ECONOMETRICS APPROACH EVALUATING THE HEDGING CAPACITY OF EX-ANTE ESTIMATIONS OF VOLATILITY

Risk control is one of the cornerstones of investment management. Theories developed in this field may directly impact the welfare of investors and therefore produce unequivocal social impact. Among the many instruments that have been studied and used as hedging instruments for portfolios, one special promise resides in the utilization of volatility instruments. From a theoretical point of view, these sorts of instruments present (the good properties of) a negative correlation with the market, return to the mean profile, and positive skewness. In such a landscape, equity portfolios could be hedged against tail risks or even be neutralized to market volatility keeping uniquely an alpha return component.

Normally, this volatility gauge is based on the ex-ante estimation of volatility embedded in the prices of the options on the equity index. The most used index in this scenario is the VIX, captured as the yardstick of 1-month expected volatility on the S&P 500. Trading on the index itself is unpractical, to not say economically impossible, as it would require an extensive portfolio of illiquid options on the index requiring burdensome rebalancing and rolling. An alternative, then, portfolio managers can use derivatives on the index, both options but mostly futures, that are easily available on the market or nowadays even the very convenient ETFs available on VIX futures.

The article evaluates the peg of VIX futures to the index, as a form to contextualize the practical efficiency of such instruments compared to the theoretical benefits of the index. It is especially important in the context that the pricing mechanism of futures contracts is not subjected to the constraints of arbitrage, considering that one cannot establish a long or short position in the underlying.

In doing so, the work provides a framework with theoretical and professional interest. Initially, it contributes to the Cost of Carry theory and the pricing of futures in general. Secondly, it contributes to the theoretical landscape of evaluating volatility as an ex-ante estimation. Lastly, it depicts the real expectations that investment managers should have regarding the usage of volatility instruments for hedging or speculation, and draws on the operationalization of the mechanics of this utilization.

The title of the article, presented in chapter 2, is *Arbitrage Implications in Pricing of Volatility Futures.*

### 1.1.2. ARTICLE 2: A HIGH-FREQUENCY APPROACH TO UNDERSTANDING VOLATILITY IDIOSYNCRASIES OF THE SIZE FACTOR IN EQUITY PORTFOLIOS

After the work of Fama and French, factor analysis became an important topic in equity investing. The inclusion of additional risk factors to the market factor prescribed by the CAPM promises to drive a more rational portfolio construction and eventually shed light on risk premiums that used to be hidden within the single factor. Strategies such as the Smart Beta were developed to tilt portfolios towards certain risk factors to induce better portfolio performance. Among these factors, return comparison between small and large-cap companies, known as the size factor, is remarkably important and generated many indexes designed to capture a specific risk characteristic.

As there is general controversy related to the existence, or at least the persistence, of a risk premium related to small companies, it is the first time that a comparison of these time series under a high-frequency regime is made to punctuate differences in the behavior of these series, pointing to a difference in the return profile. By doing so, the article seeks to point only to the microstructure difference between small and large-caps, agreeing with the existence of a premium related to factor, but in any way, advocating a stable existence of a fixed premium.

This article is designed to generate both theoretical and practical benefits. In the first view, it contributes to the concept of volatility by establishing a bridge between the high-frequency approach and the market volatility. Additionally, the paper depicts a real application of the high-frequency theoretical framework in a practical investment problem, opening a field of work for such methods in real life. At last, a practical evaluation of the size factor is provided, as well as a framework for the evaluation of other factors, impacting directly the work of portfolio managers and investors.

The title of the article, presented in chapter 3, is *Small Firm Factor: Riskiness Evaluation Under a High-Frequency Regime.*

### 1.1.3. ARTICLE 3: EVALUATION OF CONDITIONS TO TIME SERIES DISRUPTION USING ULTRA-HIGH-FREQUENCY

The observable characteristics of a time series change as it approaches a continuous time profile. In such a situation, at least two important new features are detected. Initially, the occurrence of the events (i.e. a contract negotiated generating a market price, for example) is not evenly distributed and therefore, one can find information gaps. In the limit, the fact that there is, or not, an information point in a specific time already carries some information to the innovation process. Secondly, under such an instantaneous approach, normal time-series features as the drift becomes unobservable and the process is composed basically of noise, Brownian motion, and jumps, meaning that on its microstructure, every time-series is built only from volatility.

Using public data from the BOVESPA, and also from an agreement that Mackenzie University established with the stock exchange, the article will measure the potential jump component of equity securities on a periodicity that extends up to 1/1000 of a second. As the IBOVESPA is composed of around 90 stocks, the evaluation of jump occurrence can be compared to a set of security features, thriving to uncover potential enablers of time-series break-points.

The article aims to establish an important theoretical contribution on the grounds of volatility and time-series econometrics. Procedures nowadays used for the evaluation of this kind of problem are still under construction, and a new theoretical framework may evolve with implications for the theory in finance and the asset management industry. In this way, the work is also professionally tilted, as it may potentially indicate enablers that may enhance volatility for intraday trading and the consequential spillover to lower frequencies.

The title of the article, presented in chapter 4, is *Jump Identification and Its Enablers: An Ultra-High-Frequency Empirical Evaluation.*

## 1.2. THEORETICAL FRAMEWORK: TRADITIONAL APPROACHES

This session briefly describes the theoretical foundations used in this research. As such, it is a guide on the evolution of econometric models with the simple function of providing elements to the reader before it goes on to the articles presented in chapters 2 to 4. Due to this fact, this theoretical review must not be seen as part of any of the subsequent articles in this research.

As a model that includes observations in different periods, time series conveys important concerns with respect to the trends, variance changes, and seasonality that are specific to this kind of modeling. As a result, DeFusco et al. (2007) comment that regular linear regression models most often have violated assumptions when dealing with this kind of dataset, regarding, for example, the correlation of residuals or stationarity.

A simple forecast method, known as the Trend Model, can be employed using linear or log-linear vectors, as exemplified in Equations (1.1) and (1.2). The difference between the models consists of the exponential growth pattern captured by the log-linear model.

In this type of model, the value of a variable at time t is a direct function of time, mediated by intercept and slope coefficients, using OLS (Ordinary Least Square), and therefore, the assumptions for linear regression models should be confirmed for validity.

$$y_t = b_0 + b_1 t + \varepsilon_t \quad (1.1)$$

$$\ln y_t = b_0 + b_1 t + \varepsilon_t \quad (1.2)$$

where,

$y_t$ is the value of the time series at time t

ln $y_t$ is the natural logarithm of $y_t$ at time t

$b_0$ is an intercept

$b_1$ is a slope

$\varepsilon_t$ is the random error

A second style of time-series models includes 'Autoregressive Models (AR)', in which a time series is regressed on its own past values, as depicted by Equation (1.3). A key feature of such a model is that the independent variable $x_{t-p}$ is also a random variable, in the sense that it is covariance stationary, holding constant properties like mean and variance over time, as well as the absence of autocorrelations.

$$x_t = b_0 + \sum_{p=-1}^{-n} b_{t-p} x_{t-P} + \varepsilon_t \quad (1.3)$$

where,

$x_t$ is the value of variable x at time t

$b_0$ is an intercept coefficient

$b_{t-p}$ is a slope coefficient at time t-p

$x_{t-p}$ is the value of variable x at time t-p

$\varepsilon_t$ is the random error

Another kind of model, that can be used in conjunction with AR is the 'Moving Average (MA)' model, which is a time-series approach based on the previous residual values and different from the smoothing technique that shares this name. As depicted in Equation (1.4), an order 1 MA model utilizes a coefficient θ to mediate the independent variable or the residual in t-1.

As AR and MA are used together, ARMA models may be defined with two sets of equations that complement each other in the time-series forecast. Additionally, AR models can be calculated considering

conditional heteroskedasticity as part of the assumptions, meaning that the variance of the residuals is correlated with the independent variable or the lagged value of the time series. When such a situation occurs, the hypothesis testing of coefficients for AR, MA, and ARMA models will not be valid, as the standard errors will be incorrect.

$$x_t = \varepsilon_t + \theta \varepsilon_{t-1} \tag{1.4}$$

where,

$x_t$ is the value of variable x at time t

$\varepsilon_t$ is the random error at time t, with expected value $E(\varepsilon_t)=0$ and $E(\varepsilon_t^2)=\sigma^2$

$\varepsilon_{t-1}$ is the random error at time t-1, expected uncorrelated to $\varepsilon_t$

Engle (1982) works with a model where the residual value, at time t, is normally distributed with an average equal to 0, the variance of the residuals dependent on the lagged variance in the time series, with intercept and slope coefficients that are obtained in the model estimation, as shown in Equation (1.5), for an order 1 model.

The 'Auto-Regressive with Conditional Heteroskedasticity (ARCH)' model becomes useful as it allows for a change in forecast uncertainty as time evolves and the fact that large and small errors tend to cluster together. This characteristic may be interesting in many econometric applications such as inflation forecasts (as done in the article), where economic theory suggests that agents tend to react to the mean "but also to higher moments of economic random variables" (ENGLE, 1982, page 1000), as in the case which inflation is associated with inflation variance, inverting the negative slope relationship with unemployment suggested by Phillips curve.

$$\varepsilon_t^2 = a_0 + a_1 \varepsilon_{t-1}^2 + u_t \tag{1.5}$$

where,

$\varepsilon_t$ is the random error at time t

$\varepsilon_{t-1}$ is the random error at time t-1

$u_t$ is the error term, different from ε as it is the error in estimating the error

The usage of models such as ARCH, and its generalized peer GARCH, underlines the fact that time-series analysis may be superiorly performed when the model allows for the capability of a changing profile of variance over time. The usefulness may be directly associated with the above-mentioned variance cluster behavior, which should be dependent on the right definition of the order of the regression.

Another path would be to consider models in which a random movement is directly incorporated into it. Initial models that follow this logic are derived from 'Random Walk' and 'Brownian Motion' and may derive into models where specific continuity and rupture elements are decoupled to generate a more

complete view of the expected behavior of the time series. In the next session, we will evolve toward this kind of model.

### 1.2.1. CONTINUOUS STOCHASTIC DIFFERENTIAL EQUATION MODELS

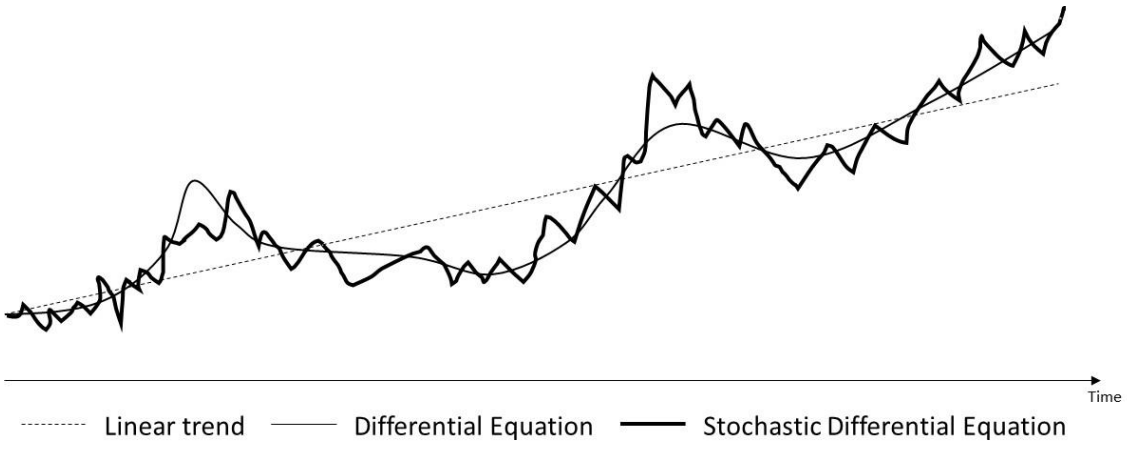
According to DeFusco et al. (2007), the models evaluated in the previous session consider a reversion pattern of the time series to its mean. Other methodologies, on the other hand, include a random noise element. In this kind of model, as defined by Evans (2013), the mathematical grounds for forecasting move to stochastic differential models, where the function pattern assumes an unsmoothed character, as shown in Figure 1.2.

Formally stating, Karatzas & Shreve (1998) define such models as a collection of random variables, based on the probability function in the Sample Space (Ω) and Event Space (F) and, therefore, measurable under the dimensional distribution of the probability and time, or progressively measurable in terms of its stopping times.

As demonstrated in Figure 1.2, Evans (2013) shows that a continuous and differentiable series can be modeled with a smoothing vector b, in such a way as explained for AR models. The equation for this kind of model would assume a form of $x_t = b\left(x_{(t)}\right)$. In the case of the stochastic differential equation, the modeling of the stochastic model, $x_t = b\left(x_{(t)}\right) + B\left(x_{(t)}\right)\varepsilon_{(t)}$ considers the inclusion of a random term to deal with the uneven pattern of the curve.

*Figure 1.2*

**Trajectory of Differential and Stochastic Models**




As shown by DeFusco et al. (2007), in its simpler case, a 'Random Walk' model, as depicted by Equation (1.6), considers that the best forecast for a specific period would be the value of the variable in the previous period plus a random number, that cannot be predicted.

$$x_t = x_{t-1} + \varepsilon_t \tag{1.6}$$

where,

$x_t$ is the value of variable x at time t

$x_{t-1}$ is the value of variable x at time t-1

$\varepsilon_t$ is the random error at time t, with $E(\varepsilon_t)=0$, $E(\varepsilon_t^2)=\sigma^2$, and $\varepsilon_t$ has no autocorrelation

Such a 'Random Walk' model has some important features. Initially, it can be seen as a special case of an AR (1) model with an intercept equal to zero and a slope equal to one, generating an undefined mean reversion pattern and a variance of errors that is directly proportional to time, and therefore, limiting to infinite as time grows. Based on these facts, normal regression analysis cannot be employed to solve the model (as it would be the case for AR models), unless the function is modified through differentiation. An extension of this discussion would reside in the understanding that this process is bounded by a 'Brownian Motion' mechanism, as explained by Au, Raj & Thurston (1997).

As a historical note, Lavenda (1985) describes the mathematical mechanism of 'Brownian Motion' since the original studies of botanist Robert Brown related to the agitation of particles in a fluid to modern applications such as economics or navigation systems. Such comparisons with economics and stock market returns are also provided by Ait-Sahalia & Jacod (2012).

This concept had crucial relevance in atomic physics and eventually arose into the segregation of a general drift movement and short-term random fluctuations around it. As an example, the Langevin equation, in thermodynamics, decouples movement in a macroscopic world drag force (temperature differentials, for example) and a microscopic fluctuating Brownian force.

According to Evans (2013), as the white noise is transformed to a Brownian Motion or Wiener Process $W_{(t)} = \varepsilon_{(t)}$ , a Stochastic Differential Equation is derived from the original Ordinary Differential Equation which may be solved using a stochastic integral approach, as depicted in Equations (1.7) and (1.8).

$$\begin{cases} dx_t = b\left(x_{(t)}\right)dt + B(x_{(t)})dW_{(t)} \\ x_{(0)} = x_0 \end{cases} \tag{1.7}$$

$$X_t = x_0 + \int_o^t b\left(x_{(s)}\right) ds + \int_0^t B\left(x_{(s)}\right) dW \tag{1.8}$$

This equation can be broken into three parts. The second part is a non-aleatory moment related to a drift and the third part is a stochastic movement described as Brownian Motion. In this point, three important points must be briefly discussed: how the definition of martingales aligns with this model, the application of Wiener process properties to Brownian Motion, and the interpretation of White Noise.

Starting with the notion that martingales are random processes in which the knowledge of previous moments, or stopping times, cannot provide information for future movements of the variable, Karatzas & Shreve (1998) underline that the standard example of such process is a one-dimensional Brownian Motion, that also could be directly related to the continuous time version of symmetric random walk. This is, in fact, the model explained in this session, meaning that it refers to a martingale.

In addition to that, as an important application in the field of finance, a sub martingale is an interesting delimitation of that concept as it embeds a non-negative return expectation, of $E(x_t|F) \geq x_s$ for $0 \leq s \leq t \leq \infty$. The sub martingale is subjected, in this way, to Doob-Meyer decomposition as a bounded martingale and an increasing trend, which is the base for the discrete-time stochastic integral.

Karatzas & Shreve (1998) consider an alternative for building a Brownian Motion from interpolation, when a Brownian Motion $\{B_t, F_t, t \geq 0\}$, with $0 \leq s \leq t \leq \infty$ and values $B_s$ = x and $B_t$ = z, has a mean (x + z)/2 and variance (t – s)/4. The authors highlight this approach as a continuation of the one known as the Wiener process, where $E(W_t) = 0$ and $\sigma^2(W_t) = t$, under a normal probability distribution $w_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$. Hall (1969) demonstrates that the study of martingales and certain sub martingales can be analyzed using Wiener process without (with) a drift, in a sense that they would share the same law. In this way, the application of Brownian Motion to Random Walks is theoretically sound.

Following this point, Karatzas & Shreve (1998) contextualize White Noise as the dW, in terms of "statistical communication theory" (KARATZAS & SHREVE, 1998, page 395). This term is used later in subsequent models and can be interpreted in terms of the randomness implied by time.

Brownian Motion's logic is based on the probability that a certain function path will be encompassed under a certain interval at a specific point in time. The Brownian Motion follows a Normal Distribution where the expected mean is zero, indicating an unbiased path $W_{(t)} = N(0, t)$.

At this point, it becomes important to evaluate some of the aspects related to the estimation of probability density function as its implications on the Brownian Motion. As described by Parzen (1962), this problem is similar to the estimation of spectral density function on a stationary time series, so the methods used may be similar to the estimation of a maximum likelihood parameter. This may be particularly important to the objective of this paper as the central problem resides in the forecast of time series with a spectrum of variation defined by a probability distribution function derived from a specific density.

## 1.3. ALTERNATIVE MODEL BASED ON HIGH-FREQUENCY

The models presented so far encompass the hard core of finance. Most of the theory produced utilizes these models, or at least assumptions that are aligned with them. In general terms, common features include the measurement of quantities, generally risk and return metrics, that are used as inputs on models aiming at portfolio optimization or risk control. Additionally, most models treat quantities as a stable input, resulting in predicted relationship among assets that are subjected to scrutiny under a certain model.

Markowitz portfolio theory, for example, establishes a relationship among assets based on pair-wise covariances resulting in an optimization algorithm that maximizes expected return or minimizes total riskiness.

There are several issues, in real life, with this kind of approach. Initially, one cannot rely on stable measures for risk and return parameters, once they are contingent on to market itself. In general terms, it means that covariances may be severely affected by market stress. Continuing in the Markowitz example, it may directly impact the capacity of the portfolio to get protection from diversification.

Moreover, the market is a dynamic construct with the changing sentiments of buyers and sellers being computed continuously, as observed by Rupande, Muguto & Muzindutsi (2019) in the South African equity market. As news is processed, for example, the market may show a peculiar movement that traditional econometric models will fail to perceive, as a result of its sample requirement and the generation of results of average behavior, in a situation described by Fisher & Statman (2019) in which sentiment is not homogeneous among market participants.

On top of the mathematical framework, computing development allows for models with an amount of data that would be considered cumbersome in older models. One of the developments can be understood as related to high-frequency information, allowed by stock exchanges. In this sense, open and close positions, bid-ask spreads, volumes, and other relevant information can be provided in the blink of a second to traders (in the professional world) and scholars (to use this brand-new universe to redefine finance theory in terms of better portfolios with more controlled risks).

As a result, this research project can be considered new in the sense that its development is only possible under the more advanced computing capability available nowadays. Nevertheless, the traditional approach to statistics cannot be disregarded in terms of the robustness and soundness of its methodological framework and practical results.

A model specially designed for high-frequency information, developed by Ait-Sahalia & Jacod (2012) will be used to evaluate the microstructure of the Brazilian stock market. This understanding, in this regard, be seen as a complementary view of the market trend and volatility instead of a competing view with traditional models. As such, the conclusions and models pertaining to the field of finance keep their relevance and may be enhanced by the usage of high-frequency data, in regards to instantaneous volatility spikes resulting from jumps in the time series, with ‘fine tuning’ implications potentially on derivative pricing, portfolio management, risk control, and trading strategies, among others.

In this way, traditional models based on a drift, such as Markowitz Portfolio Theory, or eventually Brownian Motion, such as Black-Scholes-Merton Option Pricing, keep their central position in finance theory and market practice.

### 1.3.1. JUMPS, MARTINGALES, “BROKEN CURVES”: MODELS FOR HIGH FREQ DATA

The inclusion of a ‘Brownian Motion’ into a time series such as a random walk implies that part of the movement is aleatorily defined under a normality assumption with variance as a function of time. This

randomness, on the other hand, doesn't disrupt the continuous character of the time sample path, as explained by Ait-Sahalia & Jacod (2012).

Another set of models, as developed by the authors, would be depicted in Equation (1.9). In this case, besides the continuous random element, the model also incorporates discontinuous jumps, which would be apprehensible when returns are analyzed on a high-frequency pattern. The potential analogy, on this point, would be to the previously mentioned study of light, conducted based on spectrum changes occurring in very short time frames.

$$x_t = x_0 + \int_0^t b_s \, d_s + \int_0^t \sigma_s \, dw_s + JUMPS \qquad (1.9)$$

where the Log of Price $x_t$ results from a drift component (first part of the equation), a continuous 'Brownian Motion' (second part of the equation), and jumps.

Initially, it is straightforward that Equation (1.9) is the same as Equation (1.8), plus the jump component, which marks the discontinuity feature of the model. In effect, this kind of analysis becomes more and more relevant as the time interval in the data series reduces as a result of the usage of high-frequency data. The main reason for this is related to the fact that it allows for the presence of a finite number of large jumps in a defined time and a finite or infinite number of small jumps in the same time frame. In economic terms, the latter could be related to short-term price noises or liquidity issues while the first would be related to relevant events such as defaults or macroeconomic shocks. According to the authors, the total return of an asset is, then, the aggregation of partial returns of its components.

In this way, the Semi-martingales assumption, as generic time-series models that can be decomposed between a drift (local martingale) and a jump, can be also sub-decomposed in small jumps and big jumps. In addition to that, the continuous part can also be calculated using a stochastic volatility process and may be itself a semi-martingale. For this continuous part, the introduction of high-frequency data allowed for methods to forecast volatility using estimators such as realized volatility combined with the effect of market microstructure noise.

The objective is to decide which components need to be included in the model and their magnitude and, despite the focus on the methodology, outline the economic implications of these models. As an example, such discontinued models may change current views on normal contingent claim valuation and optimal portfolio choices, as they would be dependent on price dynamics. Moreover, many high-frequency trading rely on specific components of the model, based on small profit with a "better than 50" chance that is repeated several times over very small time intervals. By using ultra-high frequency, there's a limitation related to data sampling that is available, reducing the capacity of looking into long series or illiquid assets. In addition, market microstructure noise is always a concern.

The model proposed by the authors consists of the calculation of test statistics based on the Realized Power Variation, B, as depicted in equation (1.10), with three controls being adjusted to isolate the specific components of the time series. The first component is related to the power, p, that is a number higher or equal to zero in which values above two highlight the jump component (as a result of a quadratic

variation feature of martingales). The second component is related to the truncation mechanism, $u_n$, as a form to isolate specific jump sizes. At last, Δn is the sampling frequency.

$$B(p, u_n, \Delta n) = \sum_{i=1}^{T/\Delta n} |\Delta_i^n x|^p \, 1_{\{|\Delta_i^n x| \leq u_n\}} \quad (1.10)$$

The four tests performed, then, are:

Test to detect the presence of jumps: calculated based on the untruncated form B(p, ∞, Δn) with power higher than 2 and compared in two different sampling frequencies. As this test, $S_j$, tends to one in its limit, the presence of jumps is detected, while as it tends to zero based on a decay rate of $k^{p/2-1}$, no jumps are present and the series is based on a continuous Brownian process

Test to detect if jumps are finite or infinite: calculated using the original truncated format of equation (1.10), with power larger than 2 at two different frequencies. As this test, $S_{FA}$, tends to one in its limit, jumps are considered infinite while as it tends to the rate of $k^{p/2-1}$, jumps are finite

Test to detect if there is a Brownian Motion: calculated using the original truncated format of equation (1.10), with power shorter than 2 at two different frequencies, with k≤2. This test is, maybe, less relevant as the authors point out that it doesn't exist a non-stochastic process in the field of finance

Also, it is possible to calculate the relative magnitude of the components of the series.

# 2. ARTICLE 1: ARBITRAGE IMPLICATIONS IN PRICING OF VOLATILITY FUTURES

## 2.1. ARTICLE ABSTRACT

The paper compares the pricing mechanism for volatility (VIX) Futures considering differences related to arbitrage potential. As the VIX index is not a tradable asset, investors cannot arbitrate prices using long/short positions in the index and Futures of the index. As a result, a relevant distinction in price dynamicity is found, confirming the theoretical importance of arbitrage in Futures pricing. The study confirms the relevance of the arbitrage mechanism to volatility Futures and identifies negative carrying yield. The main implication for portfolio managers and investors is related to a potential financial burden by holding long positions in volatility Futures, especially in the mid and long terms. As such, these instruments would be better suited for short-term hedging strategies.

## 2.2. ARTICLE KEY-WORDS

Volatility, Futures Contracts, Arbitrage, VIX, Hedging

## 2.3. INTRODUCTION

Risk control is a central piece in equity investing. This notion derives from the acknowledgment that increased expected portfolio utility may be the result of the combination of higher returns and lower volatility. Under this assumption, portfolio management includes a range of strategies focused on properly measuring exposures, rationally controlling them, and optimizing performance considering both risk and return profiles. One example is related to the diversification approach, in which the combination of low-correlation stocks yields a benefit on the risk-return profile of a portfolio, following the theoretical field developed by Markowitz.

In recent years, portfolio risk management became even more prominent as a result of many shocks in the stock markets, both from an exogenous nature, such as the abnormal event related to the Covid pandemic in 2020, and a more cyclical profile, as in the case of potential monetary stance shifts related to the business cycle and macroeconomic conditions. In such cases, strategies like diversification may not prevent drawbacks in a portfolio, and more active strategies were highlighted as a source of value to investment holders and managers. Among these policies, hedging is a natural method to be evaluated.

Among several potential hedging designs, the utilization of volatility derivatives receives special attention. It conveys the idea that it may allow portfolio managers to decouple the directional results from the change in market volatility, implying a potential increase in market stress can be neutralized in portfolio returns. Normally, the implementation of such a program would require the usage of derivatives of VIX.

Nowadays, small portfolio managers may implement programs using ETFs backed by VIX futures. Trading on such instruments has dramatically increased in the past years.

As these instruments are to be used as a portfolio hedge, it is crucial then to understand their pricing logic, in order to capture the risk mitigation effects. Since the contributions of Modigliani and Miller, Black, Scholes and Merton, and Brennan in the second half of the 20th century, the traditional evaluation of derivatives and their pricing models have been developed under the framework of risk neutrality hedge positioning, and arbitrage mechanisms. This methodology presents several benefits in valuing derivatives including the fact that it does not necessitate an estimation of the required rate of return based on unobservable variables, such as the risk tolerance of individuals, which is crucial for an industrialized calculation of such instruments.

The structural logic of the valuation process, in this sense, requires the construction of hedged portfolios including long and short positions in the derivatives and the underlying assets. Under this assumption, the resulting hedged portfolio assumes a risk-free profile that earns, for this reason, a risk-free rate as a return. In every situation where prices deviate from the framework, an arbitrage opportunity would arise, in a converging mechanism that aligns prices back to the model.

In regards to Futures markets, the non-contingent cash flow nature implies a model that, as presented by Cornell and French (1983), is based on the time to maturity and the risk-free rate, known as the Cost of Carry. In such a model, futures prices represent rather an arbitrage-free condition than any future expectation of prices of the underlying, as a result of, in case of price deviation to the model, the possibility of long (short) and short (long) positions in the underlying and Futures, with a loan (investment) at risk-free rate locking into profit with no required initial cash flow from the investor.

One key element for this pricing model, and the arbitrage condition embedded in it, is the tradable nature of both the underlying and the futures, as the case of the S&P 500 Index discussed by the authors. Several scholars have analyzed different Futures markets under this framework. Bialkowski & Perera (2018) conclude that the inconclusive presence of mispricing in 36 Cost of Carry published studies. They point, though, to the main factors that are attributable to mispricing such as taxes, dividend uncertainties, unequal rates, problems with underlying, transaction costs, and short-selling availability.

This study focuses on the specific case of volatility futures, where trading is available only for the derivative but not for the underlying. More specifically, the study analyses the pricing dynamics of VIX futures and its relationship with the VIX index, by evaluating the adherence to the Cost of Carry model to a real data regression on the returns of the VIX index and VIX futures indexes. The paper aims to contribute to the extensive literature covering the application of volatility derivatives on portfolios.

This research is relevant to scholars and practitioners. Initially, it depicts the relevance of the arbitrage mechanism on the Cost of Carry pricing model and states theoretical differences in cases in which arbitrage is possible or not to be executed. Additionally, as a result of the availability of VIX Futures and ETFs based on them, and the appeal volatility to risk management as a result of asymmetric negative correlation to the stock market, an increased understanding of these instruments is crucial to the proper evaluation of risk/return profiles of such strategies.

This study proceeds as follows: it begins with a brief discussion of the arbitrage mechanism and the Cost of Carry pricing model. The third session depicts the VIX index and its derivative products on it. The fourth

session introduces the methodology and econometric model used in this study, followed by results and conclusions.

## 2.4. THE ARBITRAGE MECHANISM AND IMPLICATIONS ON VALUATION

The arbitrage-free assumption, by which many securities are valued, implies that prices are consistent with a model that eliminates arbitrage opportunities. In the eventuality that arbitrage opportunities are present, the fact that economic agents act on it triggers market correction towards the arbitrage-free situation. Arbitrage mechanisms were explored by Modigliani & Miller (1958) in defining their first theorem on capital structure irrelevance. According to their view, a mismatch in the price of stocks and bonds would allow investors to take long/short positions that would be advantageous independently of attitudes towards risk and require no initial cash flow. When opportunities like this occur, the fact that investors exploit it creates the mechanism to force convergence as arbitrage opportunity is eliminated.

Bender (2012) depicts how the absence of arbitrage opportunities is a central piece of pricing models. As prices deviate from this condition, both 'obvious arbitrage', in which a minimum riskless return is obtained, and '0-admissable arbitrage', in which no debt is required for a strategy that can wait for a positive result, is possible. This second case is remarkably interesting as stochastic components are added to the model. As admissible strategies are restricted or market friction is introduced, resulting in the reduction of the liberty to execute the strategy, the author describes disturbance in simple arbitrage opportunities.

The classification used by Bender (2012) differentiates arbitrage on the basis of its execution strategy. In regards to the nature of the opportunity, arbitrage can be seen as a result of Value Additivity, in which an arbitrage opportunity may arise as a mismatch between the price of 'whole' security and the prices of its components, and Dominance, that is a result of risk/return profiles of securities.

Several empirical tests were conducted to evaluate the effect of arbitrage. Neal (1996), for example, depicts mean reversion, for the S&P500, of mispricing as a result of arbitrage. In addition, it shows that potential mispricing on the pair Spot/Futures may disappear even before arbitrageurs can act on it. On the other hand, the author also pointed to the higher-than-expected Futures volatility, in comparison with index volatility, in disagreement with arbitrage models. Zang & Watada (2019) identify only infrequent arbitrage opportunities in the Chinese options market, generally related to low liquidity and infrequent trading. Lutkebohmert & Sester (2021) evaluates opportunities for statistical arbitrage and depicts conditions and expected performance of arbitrage strategies.

As for the effect of arbitrage in the valuation of Futures, the Cost of Carry model considers that prices adjust to eliminate arbitrage potential, due to the nature of arbitrage regarding the absence of initial cash flows and the riskless returns. As opportunities arise, market agents will explore them by constraining prices back to arbitrage conditions. As such, an understanding of the model is provided in the next session.

## 2.5. FUTURES PRICING MODEL

Forward commitments are derivative contracts that allow for a future transaction, either long or short, at a predetermined price. Under this description, both Futures and Forward contracts are included, with a difference related to the over-the-counter nature of the last. As a result, Futures contracts, negotiated on exchanges, differ from Forwards due to their standardized structure, and the mechanism of margin accounts, as per avoidance of defaults. Typically, Forward contracts are simpler, as they require only one exchange of cash and goods at maturity, and the valuation models developed for these derivatives are based on them. As Chow, MacAleer & Sequeira (2000) point to the fact that potential price differences are not economically significant, the models used in this paper will be used for Futures contracts. A potential limitation of this approach is related to assets where the interest rate is correlated to the price of the underlying. As an example, Dezhbakhsh (1994) finds statistical differences between the two contracts on the exchange rate market, due to mark to the market mechanism.

Two main theoretical frameworks were developed to evaluate the pricing of Futures and Forward contracts. The Risk Premium hypothesis implies that Futures securities could be analyzed in the same way as 'spot securities', in terms of having a risk and return profile that could be optimized in a portfolio, and view Futures markets as a sort of forecast mechanism for future spot prices.

Derived from the work of Keynes, the Risk Premium framework, according to Dusak (1973), concludes that these contracts should be in normal backwardation as a result of the insurance nature of the payoffs, where hedgers and speculators hold respectively short and long positions. Empirical results fail to confirm that risk premiums are captured by speculators and that normal backwardation captures the behavior of future spot prices.

The second hypothesis is the Cost of Carry approach. Under this framework, Cornell and French (1983) emphasize the fact that Futures pricing is constrained by the payoff of a portfolio of the underlying and the derivative in a risk-neutral hedged scenario. Chow, MacAleer & Sequeira (2000) depict that price dynamics for Futures markets will depend on the interest component, the warehousing cost, and a convenience yield in a scenario where price deviations may be immediately corrected by arbitrageurs.

The Cost of Carry embeds four main assumptions, related to the absence of transaction costs and taxes, the validity of the risk-free rate in lending and borrowing transactions, the tradability of both the derivative and the underlying, and the exploration of any arbitrage opportunities as they occur. A potential fifth assumption may be related to the availability of short selling, which may be irrelevant in the case of investment assets. In this sense, a pricing model is generated as the derivation of two positions. In the long part, the investor buys the underlying using proceeds from a loan. This operation results in a future cash flow depicted in Equation (2.1), as the investor sells the underlying and pays the loan, or the future value (FV) of the spot price at a risk-free rate. In the short part, the investor enters a short Futures contract with value according to Equation (2.2). Both long and short positions settle at the same time and, for the non-arbitrage assumption to hold, Equation (2.1) and Equation (2.2) have the same value on the settlement date.

$$LONG\ POSITION = +S_t - FV\ S_0 \quad (2.1)$$

$$SHORT\ POSITION = F_0(t)\ -\ S_t \quad (2.2)$$

In addition, as no initial investment is required, this operation can be replicated infinitely until the opportunity no longer exists, as an effect of increasing demand for the asset and supply on Futures.

As a consequence, according to Cornell and French (1983), for any investment asset that provides no income, has no storage cost, and when a short sale is possible, the pricing calculation of Futures is based on Equation (2.3). As expected, Equation (2.3) equals the equality stated for Equations (2.1) and (2.2), where the risk-free rate is continuously compounded.

$$F_0 = S_0 e^{rt} \quad (2.3)$$

where F0 is the Futures price at time 0, S0 is the spot price at time 0, r is the continuously compounded risk-free rate, and t is the time of expiration of the Futures contract.

This model applies to several markets. Sarno & Valente (2000), for example, describe a consistent long-time pattern, according to the Cost of Carry framework, considering that the mean reversion of the basis is characterized by a nonlinear dynamic, due to transaction costs, trade frequency, and regime shifts. In this way, the cointegration of the underlying and the derivative may be temporarily imperfect.

As for Equation (2.3) to be stable, the dynamics of the underlying and the Futures should be consistent. LaFuente & Novales (2003) describe three conditions in which this assumption is to be understood. Initially, it is important to notice that there is perfect co-movement between the underlying and the derivative. In this way, the models call for a perfect correlation of returns for the underlying asset and the derivative and equality of variance of returns. At last, returns are not serially correlated. The authors work with a hedge ratio calculation for a stock index that includes a stochastic noise element on top of these assumptions. Under the theoretical circumstances, nevertheless, Equation (2.4) depicts that Futures price change is a function of the return on the underlying, as stated by the authors, but also on a potential variation on the risk-free rate and the time decay to expiration.

$$F_t \partial_T = S_t \partial_T e^{r \partial_T \Delta T} \quad (2.4)$$

where the change in the Futures prices in time T towards time t is a function of the same change in the underlying and the time decay to the expiration of the contract.

As this pricing model is expected to be meaningful for Futures contracts in any sort of underlying, the attention is shifted to Futures on volatility, a specific asset class that can be used as hedging instruments or directional investment strategies. Initially, a brief discussion on the nature of volatility and its indices will be taken, so a model for empirical analysis can be built.

## 2.6. MARKET VOLATILITY, VOLATILITY INDICES, AND VOLATILITY FUTURES INDEXES

In 2003, CBOE launched the VIX index, the 30-day expected volatility for the S&P500 utilizing standard SPX options that expire every 3rd Friday of the month along with weekly options that expire every Friday, aimed at solving the issues identified in the former VXO index. The VIX methodology lies on the Volatility Swap Rates, instead of BS implicit rate, improving features related to economic meaning, investment capacity, and elimination of adjustment biases.

As an individual asset, the volatility, or more specifically the VIX, can add some benefits to portfolios regarding hedging capabilities and directional positioning. In general, these securities should be able to isolate returns from volatility dynamics, detached from the directional change in the underlying index or security. In regards to these features, Berkowitz & DeLisle (2018) point to the negative correlation between the VIX index with the S&P500, especially in moments of declining S&P500. This is called asymmetric volatility. Also, the authors describe volatility as a mean reversal and stationary process. Whaley (2009) points out that this asymmetry means the VIX works as a better barometer of fear than excitement in stock markets.

Dupoyet, Daigler & Chen (2010) acknowledge the advantages of the VIX as it uses out-of-the-money options, and therefore considers all the range of the volatility smile. Based on this, the VIX "theoretically" generates an asset that could be hedged using a combination of options. On this point, Carr & Wu (2006) note that the VIX methodology, as a replica of variance swaps based on an actual portfolio of options, allows for greater economic substance and product-related interest.

On the other hand, one critical element of this analysis is that, despite the theoretical possibility, the VIX is not traded on the spot market and may not be replicated in practice. Buetow & Henderson (2016) point to the fact that, despite the above-mentioned logic of the calculation that could imply the potential use of a portfolio of options to replicate pay-offs, many of the options included trade very infrequently and the generation of a hedging portfolio that mimics the returns of VIX is impractical. Berkowitz & DeLisle (2018) add to this fact by acknowledging that replicating the VIX would be a cost-prohibitive strategy due to the necessity of continuous trading, and the lack of available options in suitable term and strike prices.

VIX Futures often provide an alternative for investors looking for the specific payoff provided by VIX. Berkowitz & DeLisle (2018) note that VIX Futures started trading in 2004. Bialkowski, Dang & Wei (2016) observe a yearly average growth of 89% in VIX Futures trading volume from 2009 to 2015. Nowadays, these instruments are trading at unprecedented levels, indicating the increasing importance of this strategy for portfolio managers. As an example, Bloomberg (2020) reports an Average Daily Volume of 377,718 contracts in February 2020, a yearly increase of 89%. With the pandemic, VIX Futures may have raised its volumes again.

A position in VIX Futures, on the other hand, cannot be mistreated as a position in the VIX index. Simon & Campasano (2014) point to the low forecast ability, of VIX Futures, in regards to the index. As a result, expected returns on the VIX index shall be considered decoupled from the ones in the Futures. The authors, in addition, support trading strategies of shorting (longing) the VIX Futures when it is sufficiently

in contango (backwardation) with a hedge on the S&P 500 index. In this case, the hedge ratio is calculated based on the Futures itself and not the “pure” index.

In the market, a set of VIX Futures Indices were created based on such securities to provide benchmarks on these return profiles. S&P Dow Jones Indices (SPDJI) provides a family of indexes mimicking long and/or short positions in VIX Futures contracts with different maturities, returning different levels of term structure convexity, and isolating specific Vega exposures.

As for two of the most important indexes in the family, the S&P500 VIX Short-Term Index MCAP and the S&P500 Mid-Term Index MCAP are rolled each day, from the nearest term contract into the longest one, to keep constant maturity. Components of return include VIX spot movements, roll yield cost, and collateral interest (SPDJI, 2021 a / b). Table 2.1 shows the maturities of VIX Futures Indexes, including the underlying contracts embedded in each index and the rolling methodology.

In terms of the usage of the Cost of Carry model, previously depicted in Equation (2.3), the fact that Futures are rolled each day, to reflect the extension of underlying contract maturities to the index mandate, implies the elimination of time decay. Only after the whole period is rolled over, based on the Futures contract settlement date, it can be said that the index refreshed all components into the new settlement date. Anyway, time to expiry has a constant maturity profile.

Moreover, volatility Futures contracts also hold another important feature to be understood when dealing with them. Initially, Shu & Zhang (2012) found evidence that both spot and Futures volatility markets react to information and may be seen as efficient. On the other hand, as depicted by Pavlova & Daigler (2008), the convergence of VIX Futures to VIX Spot is not typical due to differences in the settlement procedure. While the spot VIX is settled based on the bid-ask average for each of the underlying S&P500 options, VIX futures rolls at the actual option price at the open quotes following the last day of Futures trading. This difference results in an average difference and standard deviation of 26 bps and 43 bps, respectively, between Spot and Futures.

Buetow & Henderson (2016) show, on the same line, that Futures basis are often large and volatile. Another important feature pointed out by the authors is the fact that VIX Futures are often in contango, generating potential impact for investors with a long position in volatility, even for hedging purposes. Simon & Campasano (2014) also concludes the prevalence of the contango situation for VIX Futures. Huang, Tong & Wang (2018) provide a pricing model for VIX Futures considering stochasticity and time-series jumps. In their model, the basis could be interpreted as the moment-generating function of the conditional expectation of the VIX index. On the same line, Yin, Bian & Wang (2021) claim superior pricing ability for VIX Futures of heterogeneous autoregressive models.

As a final point on volatility and VIX, besides large and sophisticated investors and portfolio managers, who have direct access to VIX derivatives, also retail investors can use these instruments as part of their portfolio strategies. This is the result of the several ETFs that have been launched to track volatility indexes. There are six ETFs tracking the S&P500 VIX Short-Term Index MCAP and the S&P500 Mid-Term Index MCAP, the two most tracked indexes, according to Bialkowski, Dang & Wei (2016). The most important provider in this market is Proshares. Among the ETFs available, the highest and second AUM is observed on Proshares Ultra VIX Short-Term Futures ETF (1.57 BUSD) and iPath S&P500 VIX Short-Term Futures ETF (806 MUSD), tracking the S&P 500 VIX Short- Term Futures Index TR. The typical expense ratio

on these products is in the range of 0.87% to 0.89% for 1x long ETFs and may reach 1.65% for inverse or leveraged ETFs.

*Table 2.1*

**Maturities of Option Contracts in VIX Futures Index (S&P Global)**

| Index | Underlying Contract | Roll Out (m) | Roll In (m) |
|---|---|---|---|
| S&P 500 VIX Short-Term Futures Index | 1st, 2nd | 1st | 2nd |
| S&P 500 VIX 2M Futures Index | 2nd, 3rd | 2nd | 3rd |
| S&P 500 VIX 3M Futures Index | 3rd, 4th | 3rd | 4th |
| S&P 500 VIX 4M Futures Index | 4th, 5th | 4th | 5th |
| S&P 500 VIX Short-Term Futures Index | 4th, 5th, 6th, 7th | 4th | 7th |
| S&P 500 VIX 6M Futures Index | 5th, 6th, 7th, 8th | 5th | 8th |
| S&P 500 VIX Front Month Futures Index | 1st | 1st | 2nd |
| S&P 500 Constant Vega (3%) and (6%) VIX Short-Term Futures Index | 1st, 2nd | 1st | 2nd |

Source: SPDJI, 2021c, p. 4

According to Bialkowski, Dang & Wei (2016), the first ETF on VIX was launched in 2009 by iPath and gained market due to low cost, tax efficiency, and stock-like features. The authors point to the fact that, to track Futures indexes, an ETF must roll its positions in the same fashion as the benchmarks. In addition, VIX ETNs may have a payoff generated through a swap instrument, instead of long/short positions of ETFs. This feature can add flexibility to the management of ETN products.

## 2.7. METHODOLOGY AND DATABASE

This paper focuses on understanding the dynamics of pricing of volatility Futures in terms of adherence to the Cost of Carry pricing model and its stochasticity. In these terms, two different pairs will be calculated under the Cost of Carry pricing models. In the first case, the VIX index is used as the spot volatility whereas the short-term VIX Futures is the dependent variable. The second equation relates the mid-term VIX Futures, as the dependent variable, to the short-term VIX Futures as the independent variable. As stated above, in the first equation one of the variables (VIX index) is not tradable while both VIX Futures (short and mid-terms) are tradable, even for the retail investor. The hypothesis assumed is that there will be a

difference in the pricing of short-term Futures, in regards to non-tradable spot VIX, and mid-term Futures, in regards to tradable short-term Futures. The first assumption is based on the fact that the arbitrage mechanism is broken, and Futures pricing should deviate from standard Futures pricing.

Also, another hypothesis is that the stochastic component of the pricing will not allow for simple arbitrage when volatility spikes. Such strategies could be understood as a simple short position in VIX Futures after a volatility increase or a long position after a reduction. In such a case, it could be observed a short-term reversion to the mean. The model, then, considers a stochastic component, based on a Brownian Motion dynamic, which is based on a normal distribution where $W_t = N\left(\mu_{t-1} - {}^{1}/_{2}\,\sigma^2, \sigma\right)$, considering the impact of volatility in the compounded return. For each point in time, the z-score of the position is calculated considering 30 mean and standard deviation on the returns of VIX Futures.

As a result, in order to capture these hypotheses, Equation (2.5) derives a pricing model, based on Equation (2.4), with a constant expiry period and a stochastic component derived from the basis risk and the non-arbitrage situation.

$$F_t \partial_T = \theta_1 S_t \partial_T e^{r\partial_T} + \theta_2 Z(W_{t-1}) \tag{2.5}$$

where $\partial_T$ stands for the time change that produces an instant change in the price of spot VIX and the risk-free rate, $\theta_1$,$\theta_2$ are coefficients to be determined, $Z(W_{t-1})$ is the implicit Z-score on the last period Brownian Motion.

As a result, the operationalization of the model in Equation (2.5) includes the calculation of two linear regressions that will be used to identify economic conclusions for futures pricing, with distinct arbitrage potential. Both regressions have robust standard errors to avoid the presence of heteroskedasticity, as prescribed by DeFusco et al (2007). The first regression will use the adjusted daily returns of CBOE VIX and the z-score of the prior period Brownian Motion as independent variables and the S&P VIX Short-Term Index as the dependent variable. The second model will use the adjusted daily returns of the S&P VIX Short-Term Index and the z-score of the prior period Brownian Motion as independent variables and the S&P VIX Mid-Term Index as dependent variables.

As for the abovementioned adjustment, the model follows Equations (2.6) and (2.7).

$$F' = \left((1 + F) \cdot r_f'\right) - 1 \tag{2.6}$$

$$r_f' = \Delta r_{f.}(N \,/\, 12) \tag{2.7}$$

where F' is the adjusted Futures to the change in the risk-free rate constricted by the time difference between the lower band of the Futures index and the Spot or the shorter Futures index, depending on the equation (rf).

The database used consists of 10 years of daily changes (2531 days) from March/01/2011 to March/19/2021. The series used includes the CBOE VIX, the S&P500 VIX Short-Term Index MCAP, the

S&P500 Mid-Term Index MCAP, and the 1-Year Treasury Constant Maturity Rate. The choice for these indexes resides in the importance they have in financial engineering and hedging. Even for retail investors, many investment products available track these Futures indexes. All calculations were done on Stata BE 17.0.

## 2.8. ECONOMETRIC MODEL AND RESULTS

The specified regression results are depicted in Table 2.2.

There are three main points to observe as the evaluation of the model is presented. The first point relates to the impact of the lack of arbitrage in the underlying, the VIX index. The second point is related to the effectiveness of naïve arbitrage strategies. Finally, the third conclusion arises from the convergence of the pricing model and the basis risk.

As expected, a relevant difference is observed between Regressions 1 and 2. The estimation of Regression 1 indicates the statistical significance of the constant term of the equation and the slope related to the adjusted return of the spot VIX. In Regression 2, on the other hand, the constant term fails to present statistical significance, while the slope term presented results in line with Regression 1.

As for the first economic implication, in terms of the constant coefficient, the fact that there is significance found in Regression 1 implies that there is a constant return on the VIX short-term Futures. As stated before, the arbitrage absence assumption considers that the constant return is not statistically significant. This result derives from the fact that no possible arbitrage can be done in a pair with the spot index non-tradable, confirming the flaw in the arbitrage mechanism implied in the mispricing of Futures, as predicted in this study.

In addition, in both regressions, the hypothesis that naïve strategies resulting from Brownian Motion may derive alpha is refuted in both Regressions. In this case, there is no systemic arbitrage opportunity arising from trading VIX Futures in a mean reversal logic. In this sense, it confirms the non-arbitrage environment and the functioning of pricing mechanisms.

At last, even though, in both cases, R2 is high and the F-Test is significant, it's clear that only approximately half of the Futures return derives from the spot or the shorter Futures. It may imply a changing sensitivity of volatility to time, meaning that the volatility surface is dynamic. Moreover, we corroborate studies, as the case of Pavlova & Daigler (2008), that point to imperfect convergence of Futures to Spot in volatility Futures, among other things due to basis risk.

The economic implications related to these results are relevant regarding the academic and portfolio management implications. Initially, it confirms the relevance of arbitrage mechanisms in pricing Futures. The statistical relevance of the constant coefficient implies a rupture with arbitrage, that should eliminate constant returns. Regression 1 is exactly calculated in the non-tradable pair. Regression 2, on the other hand, is subjected to arbitrage forces. The statistical significance of the constant in Regression 1 and insignificance in Regression 2 confirms the hypothesized arbitrage mechanism.

*Table 2.2*

**Regression Results using Robust Standard Error Type**

| | Regression 1 (ST Futures vs. VIX and Z-score) | Regression 2 (MT Futures vs. ST Futures and Z-score) |
|---|---|---|
| Number of Obs | 2,501 | 2,501 |
| R-Squared | 0.8109 | 0.8320 |
| | | |
| Coefficients | | |
| VIX (P-value) | 0.5231 (0.000) | |
| ST Futures (P-value) | | 0.4113 (0.000) |
| Z-Score (P-value) | 0.0006 (0.504) | 0.0006 (0.056) |
| Constant (P-value) | -0.0032 (0.000) | 0.0000 (0.797) |

## 2.9. CONCLUSIONS

The study confirms the impact exerted by arbitrage on the pricing model of volatility Futures. The constant return found in Regression 1, therefore implying that the arbitrage mechanism is broken, occurs due to the non-tradable nature of the VIX index that prevents investors from acting on pricing mismatches. As a result, the Cost of Carry assumption does not hold. On the other hand, it is possible to notice the opposite situation when the pricing model is run with short and mid-term Futures, which are tradable directly as derivative instruments or based on ETFs.

In addition to that, the parameter on Regression 1 implies a negative carrying yield on volatility Futures. On a daily portfolio management routine, it may imply two effects. Initially, it is important to distinguish returns on Futures from returns on the VIX. In this way, any expectation or claim related to VIX returns would not be reachable by portfolio managers. It also represents a question mark for studies in which the VIX is used to decouple returns from volatility and direction in the market.

The second point refers to the potential impact of holding these securities for medium and long times. Investors may be penalized by the negative yield embedded in volatility Futures and may decide to use these securities in short-term hedging strategies. As previously pointed out, many authors found a positive basis on the VIX Futures (contango), which has a direct impact on the negative constant parameter, mostly when the VIX Futures is considered to be a poor forecast element for the VIX index and that the basis roll to maturity in the form of Futures price decreases.

Another important confirmation from this study relates to the fact that data does not confirm a positive yield from simple arbitrage strategies on naïve rules (sell when volatility is above the average or buy when it's below) which reinforces the validity of the arbitrage theoretical framework on volatility Futures.

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# 3. ARTICLE 2: SMALL FIRM FACTOR: RISKINESS EVALUATION UNDER A HIGH-FREQUENCY REGIME

## 3.1. ARTICLE ABSTRACT

Since the advent of multifactor models, the identification of factors that could lead to abnormal returns and their portfolio implications has been the theme of consistent effort and study. Among these factors, the small-cap factor (SMB factor) is one of crucial importance. This paper provides an innovative approach to the evaluation of the SMB factor taking into consideration the time-series comparison of small and large caps under a high-frequency regime, identifying structural differences in terms of stochasticity and presence of jumps. The empirical results confirm the specificity of small caps, understood therefore as a diversifying asset class. Additionally, results fail to confirm abnormal returns in the sample, indicating that factor momentum shall be considered. Moreover, the paper demonstrates a framework for high-frequency statistics usage in investment evaluation.

## 3.2. ARTICLE KEY-WORDS

Rewarded factors, small firm factor, Brownian Motion, jump, market risk

## 3.3. INTRODUCTION

Investors and portfolio managers are often concerned about the drivers of return and risk in their portfolios. As a collection of stocks, equity portfolios consist of a combination of specific characteristics related to the specific portfolio components and their combination. Initially, different firms typically embed investment ideas, expected to be profitable under an economic scenario. Moreover, the diversification feature may improve the riskiness profile. Investment strategies cannot, in this sense, be evaluated exclusively in terms of realized returns but rather in terms of the risk exposure utilized to achieve a return.

Measurement and evaluation of risk, then, is a cornerstone in financial literature. Ali & Bashir (2022) perform a bibliometric review of asset pricing that points to this relevance. One of the most important models in this direction, the CAPM, points to a positive relationship between a stock and the market risk, measured by the β coefficient. As idiosyncratic risk is diversified away, investors would retain a certain exposure to the market risk, the only source of risk. In such a model, firms are treated as stand-alone risk components of a portfolio, which is somewhat untrue in reality. Stocks and firms share underlying common risk factors among them, potentially related to specificities of economic sectors, markets in which they operate, supply chain constraints, and many other characteristics. In this way, besides the market risk, portfolio design also implies tilts toward specific factors, highlighting the need to understand these additional exposures.

Blitz & Vidojevic (2019) evaluates factor exposures in the portfolio logic. In this context, portfolios can be understood as a collection of distinct factor exposures, which may drive the capacity of tilting towards a specific directional view. Strategies in this direction, also known as smart betas, may improve the return/risk profile of portfolios.

Fama & French (1992) pioneer this relationship by the inclusion of two additional risk factors, namely the size and the value effects. The first exposure is related to the comparison of the returns of small and large market capitalization firms and the last is relative to the book-to-value ratio. In this way, the authors identify the risk of a stock as a multidimensional vector, with one key element related to the log of the market capitalization of the firm. As a consequence, an investment and portfolio management strategy may be derived, taking into consideration the special features of smaller firm equity, in terms of its risk and return profiles. Hackel, Livnat & Rai (1994) observed, indeed, that financial institutions started offering small-cap investment vehicles as soon as this research was released.

Among the theoretical reasons for such idiosyncratic performance, some articles postulate that the information differential between small and large firms would be the central piece for the distinct investment performance, as in the case of Barry & Brown (1984). In addition to this, another line of research focuses on productivity differentials and funding needs. Works in this line, as in the case of Kim & Burnie (2002), may attribute higher performance volatility to small firms as a result of higher operating and financial leverage and lower access to scale and technology, resulting in higher exposure to economic cycles.

He & Ng (1994) relate the riskiness of the size factor to distress, which could be characterized both by cash flow generation volatility and risk of failure. In addition, the size factor correlates with both yield curve term and credit spreads, indicating high sensitivity of smaller firms to the economic cycle, even considering the potential compensatory effect of a healthier (weaker) economy, with higher (lower) term premium, and the improvement (de-crease) of credit risk, reducing (increasing) credit premium. In another macroeconomic discussion, Everett & Watson (1998) acknowledge that a small firm's mortality is directly related to interest rate dynamics, besides internal causes such as lack of management skills and capital.

Babenko, Boguth & Tserlukevich (2016) identify size risk premium by segregating the effects of rewarded systematic risk, as dictated by the CAPM model, and idiosyncratic cash flow shocks. The research suggests that small firms should have higher required returns in accordance with higher risk.

In terms of empirical research, Hackel, Livnat & Rai (1994) point to abnormal returns using a portfolio of small firms with consistent operating cash flow generation. Switzer (2010) points to an average premium of 2.03% for small-cap equities, with a relevant degree of variability in time. The author, in discussing if this factor is still relevant on the market, observes the fact that the factor Jensen's Alpha regained statistical significance post-year 2000. Moreover, the finding that the performance of small firm equities is dependent on the stage of the economic cycle and country idiosyncrasies may explain such alteration of the size effect in time. On a global analysis over three decades, Hou, Karolyi & Kho (2011) don't favor the size effect, potentially indicating that the factor may not be constant and globally diffused.

Gandhi & Lustic (2015) find evidence of a size effect in the US banking industry. This factor, which is distinct from the size factor in non-financial firms, also plays a role related to riskiness in times of financial distress.

The main difference to be taken, according to the authors, is that, in the case of banks, government bailing is more probable in bigger institutions.

In general, the studies on factor investing compare the cross-sectional performance of stocks with subsequent time series modeling to understand the 'average' loading of the specific factor β or contrast the performance of portfolios tilted by a specific factor. These works can be understood as an external approach to the risk factors that seek to identify the existence of the anomaly and eventually propose reasons based on co-movement with other variables or occurrence in time and space.

This paper intends to provide an innovative view of the size factor. This research is the first, to my knowledge, that seeks to evaluate the intrinsic characteristics of distinct time series (more specifically, small and large-caps) regarding their degree of continuity and disruption, using the lenses of high-frequency data. The goal is to understand the mechanical differences in the time series of small and large cap indices, and therefore comprehend how the risk manifests itself for different factor loadings. By comparing different Russell indices in high frequency, we depict inside-out features of time series for large and small caps, pointing to their idiosyncratic features and allowing for an analysis of the risk and return profile related to the size effect. The choice for this family of indices is also an important feature of the paper as it emphasizes proper comparison, not influenced by inconsistencies in index rebalancing and re-constitution, to mention a few.

Results obtained clearly differentiate the Russell 2000 Index, the small-cap index scrutinized, from the Russell 1000 Index, the large-cap index, and Russell 3000 Index, the broad market index. Moreover, time series jumps are asymptotically more probable in the Russell 1000 Index, in the sample analyzed. As sample dependency is plausible, the remarkable difference between the small and large caps supports the theoretical approach of considering size as a specific factor exposure. Another important point aligns with the similarities of the large-cap index and the broad market one, implying in low diversification potential of the risk factor unless a specific small-cap vehicle is directly used in the portfolio. At last, all three time series demonstrate the presence of a stochastic component, following a Brownian Motion dynamic.

This article aims to contribute to the theoretical framework of finance and investments in two ways. Initially, it depicts a significant difference in the micro-structure of the time series of the small-cap index, as indicated by a different pattern of jump incidence, when compared to large firm equities. Based on this point, it allows for a distinction between the riskiness pattern of small caps. Also, the research offers a fresh view of the mechanism behind the distinct risk pattern of small caps, from an angle not yet pursued, to the extent of my knowledge. This approach, related to the high-frequency comparison of microstructure risk, is in our view not yet explored in the literature. Additionally, the research also aims to indicate points of attention and action to practitioners in investment management. A broader understanding of risk sources is a crucial tool in evaluating portfolio strategies that may make use of smart beta strategies or explicitly include the evaluation of re-warded risk factors in its composition. At last, this essay brings a practical case study using high frequency, contributing to an analytical path that is still under development.

This paper is structured in the following manner. Session 2 describes different time series models. The session follows a crescendo, in terms of features utilized in the models, culminating with the utilization of features and culminating with a model that allows for discontinuity under high-frequency data. Session 3 discusses the data used in the paper, consisting of 1 minute period time series for the three indices

evaluated. Session 4 describes the econometric model used in the research. Session 5 depicts the empirical results achieved and highlights economic implications. The work closes with concluding remarks on Session 6.

## 3.4. THEORETICAL FRAMEWORK

According to Tsay (2000), the utilization of time series in business and economics can be related to the understanding of the dynamic structure of a process including seasonality and relationship among variables, in which the presence of serial correlation improves regression estimates and allows for forecasting production. In general terms, a time series is produced as a next-period forecast of process-derived mechanics, in which the information set available is used to predict the forward point and an innovation shock that can be associated with the error, noise, or a stochastic process.

Among the many distinctions and classifications potentially used on this subject, this paper shed light on the features included in some models to capture the stochasticity of the movement around univariate time series, which could, under the high-frequency regime, imply in discontinuity of the series. The models presented are not an exhaustive list of time series methodologies existing nor are arranged in a historical timeline. The classification proposed suits the understanding of these models as a set of features included in each that would lead to a specific model design and result interpretation.

Distinct classifications, such as time domain and frequency domain approaches, traditional or Bayesian, and univariate and multivariate, among others, as well as discussions about non-linearity or non-normality and distinct models, can be found in Tsay (2000) and Macaira et. al. (2018) but are not addressed in this paper.

### 3.4.1. CONTINUOUS STOCHASTIC DIFFERENTIAL EQUATION MODELS

Continuous stochastic models include a random noise element in their construction without representing a disruption in the time series path. In this kind of model, as defined by Evans (2013), the mathematical grounds for forecasting move to stochastic differential equations, as an aleatory element is included in the function, resulting in movement uncertainty in the time series.

Formally stating, Karatzas & Shreve (1998) define such models as a collection of random variables, based on the probability function in the sample space (Ω) and event space (F) and, therefore, measurable under the dimensional distribution of the probability and time, or progressively measurable in terms of its stopping times. Evans (2013) shows that a continuous and differentiable series can be modeled with a smoothing vector b. The equation for this kind of model would be expanded from the one presented in autoregressive models to assume a form of $x_t=b(x_{(t-p)})+B(x_{(t)})\varepsilon_{(t)}$ which considers the inclusion of a random term to deal with the uneven pattern of the curve.

As shown by DeFusco et al. (2007), in its simpler case, a random walk model considers that the best forecast for a specific period would be the value of the variable in the previous period plus a random

movement, that cannot be predicted. Such a model has some important features. Initially, it can be seen as a special case of an AR (1) model with an intercept equal to zero and a slope equivalent to one, generating an undefined mean reversion pattern and a variance of errors that is directly proportional to time, and therefore, limiting to infinite as time grows. Based on these facts, normal regression analysis cannot be employed to solve the model (as would be the case for AR models), unless if the function is modified through differentiation. An extension of this discussion would reside in the understanding that this process is bounded by a Brownian motion mechanism, as explained by Au, Raj & Thurston (1997).

On a historical note, Lavenda (1985) describes the mathematical mechanism of Brownian motion since the original studies of botanist Robert Brown related to the agitation of particles in a fluid to modern applications such as economics or navigation systems. The concept had crucial relevance in atomic physics and eventually arose into the segregation of a general drift movement and short-term random fluctuations around it. As an example, the Langevin equation, in thermodynamics, decouples movement in a macroscopic world drag force (temperature differentials, for example) and a microscopic fluctuating Brownian force.

According to Evans (2013), as the white noise is transformed to a Brownian motion or Wiener process W(t)=ε(t), a Stochastic Differential Equation is derived from the original Ordinary Differential Equation, as described in the AR model, as it can be noted in Equation (3.1).

$$\begin{cases} dx_t = b\left(x_{(t)}\right)dt + B(x_{(t)})dW_{(t)} \\ x_{(0)} = x_0 \end{cases} \tag{3.1}$$

Which may be solved using the stochastic integral approach as depicted in Equation (3.2).

$$X_t = x_0 + \int_o^t b\left(x_{(s)}\right) ds + \int_0^t B\left(x_{(s)}\right) dW \tag{3.2}$$

This equation can be broken into three parts. In addition to the intercept, the second part is a non-aleatory moment related to a drift and the third part is the stochastic movement. In this point, three important issues must be briefly discussed: how the definition of martingales aligns with this model, the application of Wiener process properties to Brownian motion, and the interpretation of white noise.

Starting with the notion that martingales are random processes in which the knowledge of previous moments, or stopping times, cannot provide information for future movements of the variable, Karatzas & Shreve (1998) underline that the standard example of such process is a one-dimensional Brownian motion, that also could be directly related to the continuous time version of symmetric random walk. This is, in fact, the model explained in this session, meaning that it refers to a martingale.

In addition to that, as an important application in the field of finance, a sub-martingale is an interesting delimitation of that concept as it embeds a non-negative return expectation, of $E(x_t||F) \geq x_s$ for $0 \leq s \leq t \leq \infty$.

The sub-martingale is subjected, in this way, to Doob-Meyer decomposition as a bounded martingale and an increasing trend, which is the base for the discrete-time stochastic integral.

Karatzas & Shreve (1998) consider an alternative for building a Brownian motion from interpolation, when $\{B_t, F_t, t\geq 0\}$, with $0\leq s\leq t\leq \infty$ and values $B_s = x$ and $B_t = z$, has a mean $(x + z)/2$ and variance $(t-s)/4$. The authors highlight this approach as a continuation of the one known as the Wiener process, where $E(W_t) = 0$ and $\sigma^2(W_t) = t$, under a normal probability distribution $w_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$. Hall (1969) demonstrates that the study of martingales and certain sub-martingales can be analyzed using Wiener process and drift, in a sense that they would share the same law. In this way, the application of Brownian motion to random walks is theoretically sound.

Following this point, Karatzas & Shreve (1998) contextualize white noise as the dW, in terms of "statistical communication theory" (KARATZAS & SHREVE, 1998, page 395). This term is used later in subsequent models and can be interpreted in terms of the randomness implied by time.

Brownian motion's logic is based on the probability that a certain function path will be encompassed under a certain interval at a specific point in time. The Brownian motion follows a Normal Distribution where the expected mean is zero, indicating an unbiased path W(t)=N(0,t).

At this point, we evaluate some of the aspects related to the estimation of probability density function as its implications on the Brownian motion. As described by Parzen (1962), this problem is similar to the estimation of spectral density function on a stationary time series, so the methods used may be similar to the estimation of a maximum likelihood parameter. This may be particularly important to the objective of this paper as the central problem resides in the forecast of time series with the spectrum of variation defined by a probability distribution function derived from a specific density.

### 3.4.2. JUMPS, MARTINGALES, "BROKEN CURVES": MODELS FOR HIGH-FREQUENCY DATA

The models presented until now assume that the time series is continuous. This is true even in the case where a stochastic element is considered, such as presented in Equation (3.3). Ait-Sahalia & Jacod (2012) state that the asset price must follow a semi-martingale assumption in order to avoid arbitrage opportunities. As analyzed, the drift and the Brownian motion elements correspond to this assumption in Equation (3.3). On the other hand, the authors work with an additional component in their model, opening the possibility of curve discontinuation due to the presence of jumps.

Makinen et al. (2019) define jumps as large price movements in microstructure that will extrapolate the explainable capacity of the Brownian motion. As it creates a relevant discontinuity on the price path, the return potentially obtained in the jump is abnormal to the continuous innovation process. From another point of view, jumps impose a high degree of realized volatility, which may spark volatility spikes in the traditional time series as volatility clusters.

The evaluation of such models is becoming more feasible due to the onset of high-frequency data, pushed by developments in computing and information technologies. From the information availability standpoint, open and close positions, bid-ask spreads, volumes, and other relevant stock market information can be provided in the blink of a second to practitioners and scholars. From a trading point

of view, Makinen et al. (2019) indicate that the technological advancement of systems and algorithms enabled by enhanced computing and colocation schemes, and that may manage a great number of market orders in a split of milliseconds.

Goodhart & O'Hara (1997) highlight this development on the potential for a continuous observation of transactions, not bounded by discrete time frames. This potentially new perspective may impose crucial changes on econometric models as the process may become time-varying.

The reason for this point is that the analytical limit of this data is related to the possibility of recording each transaction as a separate time moment. In such cases, defined as ultra-high-frequency, Engle (2000) defines a model in which the information embedded in the transaction is a random variable but also its timing is. For this strongly granular approach, many moments in time will not register a single transaction, meaning that its occurrence per se implies informational interest. Moreover, as more moments are filled with transactions, the author identifies the volume as the reciprocal of duration between the trades, which is negatively correlated with moment volatility.

In addition to this point, Goodhart & O'Hara (1997) associate market movement with the number of transactions, in a way that the high-frequency data series can still be seen as a martingale but not a Markovian, meaning that the enchainment of prices does influence the next stages of the time series. In such a way, the market is a dynamic construct with changing sentiments of buyers and sellers being computed continuously, with some implications. Examples of these effects are that, as news is processed, the market may show a peculiar movement that traditional econometric models will fail to perceive or, as critical technical points are reached, volatility may be severely impacted until confirmation or rejection of the signal is clear.

In this way, a high-frequency periodicity allows for a distinctive pattern in which it may be broken in such moments of abnormal transactions. In more formal terms, previous findings of volatility clustering can be translated in terms of the number of transactions and the potential presence of jumps in the time series.

The described inclusion, in the previous session, of a Brownian motion into time series such as a random walk implies that part of the movement is aleatorily defined under a normality assumption with variance as a function of time. This randomness, on the other hand, doesn't disrupt the continuous character of the time sample path, as explained by Ait-Sahalia & Jacod (2012).

The model developed by the authors is depicted in Equation (3.3). In this case, besides the continuous random element, it also incorporates discontinuous jumps, which would be apprehensible when returns are being analyzed on a high-frequency pattern.

$$x_t = x_0 + \int_0^t b_s \, d_s + \int_0^t \sigma_s \, dw_s + JUMPS \qquad (3.3)$$

This model raised a strong interest in analyzing markets. Pan (2002) acknowledges the presence of both stochastic and jump components in stock returns and that the risk of jumps is priced into options, and therefore into implied volatility, especially in moments of higher spot volatility. Makinen et al. (2019) develop a jump prediction methodology based on the asymmetry of limit orders book. Hu et al. (2019)

identify the jumps as a source of market volatility, in an environment where a jump in stock, especially in the financial sector, may contaminate other firms.

It is straightforward that Equation (3.3) is the same as Equation (3.2), plus the jump component, which marks the discontinuity feature of the model. The application of the model proposed by Ait-Sahalia & Jacod (2012) seeks to decide which are the components to include and their magnitude, besides outlining the economic implications of these models. As an example, such discontinued models may change current views on normal contingent claim valuation and optimal portfolio choices, as they would be dependent on price dynamics. Moreover, many high-frequency trading strategies rely on specific components of the model, based on small profit with a "better than 50" chance that is repeated several times over very small time intervals. By using ultra-high-frequency, there's a limitation related to data sampling that is available, reducing the capacity of looking into long series or illiquid assets. In addition, market microstructure noise is always a concern.

## 3.5. DATA

The data utilized in this research comprises 54 trading days from Oct/10th/2021 to Jan/4th/2022. The Russell 1000 Index, Russel 2000 Index, and the Russell 3000 Index were collected based on a 1-minute periodicity, exclusive to regular operating times of stock exchanges, 9:30 AM to 4:00 PM. In all data points, the closing index is recorded, with the exception of 9:30 AM when both the closing and opening prices are recorded. The last point collected is the closing of 3:59 PM. Returns are calculated using continuous compounding from one minute to another, excluding the first minute of trading each day, as suggested by Ait-Sahalia & Jacod (2012) as a result of the distinct volatility profile of that first minute.

Each trading day in the sample contains 389 points, with the exception of Black Friday on Nov/26th/2021, where the market operated until 1:00 PM. Based on this, the database consists of 20,826 points for each of the three indices.

The Russell family of indices is one of the most important standards in the market, with more than 9 trillion USD of assets directly benchmarked. They are designed to evaluate broad market performance and allow for specific views of market subsets, in terms of size and style (FTSE, 2021). Moreover, the cohesive methodology allows for direct comparison among those indices without external problems, such as reconstitution and rebalancing. Basically, the entire family derives from an index that contains the 4,000 largest public companies in the US, from which the Russell 3000 Index is generated as a gauge for the broad market, including 3,000 firms. Both Russell 1000 and Russell 2000 Indices are subgroups of the Russell 3000 Index, where the largest 1,000 companies are included in the first as a measurement of the large-cap segment, and the smallest 2,000 firms are included in the last as a gauge of the small-cap segment.

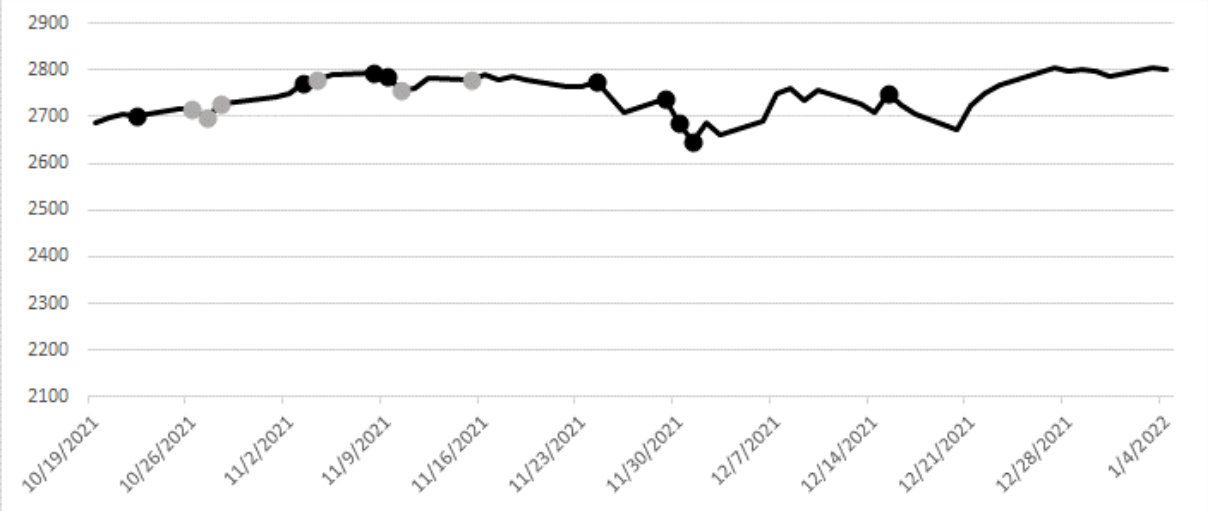
During the period, the performance of small caps largely lagged large caps, with a period return of 0.05% against 5.32%. In addition, a visual inspection of Figure 3.1 unveils a higher degree of volatility associated with the first, in the time frame. In numeric terms, the daily standard deviation of the small-cap index is 1.40% while the same statistic for the large-cap is 0.89% within the sample. Another important point depicted in the chart is related to relevant points in terms of volatility events. During the period, there

were 9 dates related to interest rate decisions (Nov/3rd/2021, and Dec/15th/2021), Fed meeting minutes publication (Nov/24th/2021), and general speeches of Fed Chairman J. Powell. All these events are marked in black. Also, 8 days were selected as the most important earnings-reporting days in the period, ranging from 217 to 660 companies reporting on the day. Those days are depicted in gray.

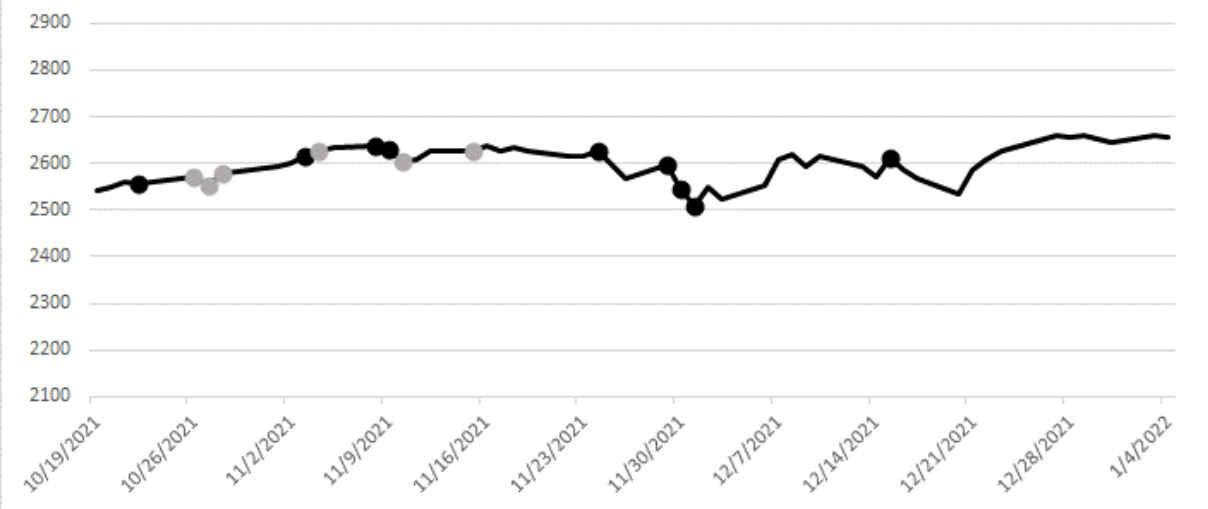
*Figure 3.1*

**Time Series of Russell Indices**

Panel A: Russel 3000 Index with Selected Dates




Panel B: Russel 1000 Index with Selected Dates



Panel C: Russel 2000 Index with Selected Dates




Source: Model Calculations

The chart demonstrates that volatility in the indices is not exclusive to the occurrence of events. By evaluating abrupt changes in the time series, it is possible to understand that some volatility clusters coincide with event days but this is not a necessary condition. It seems, though, that Fed days and busy announcement days are important for the market. In addition, the change in slope on the Russell 2000 Index has a more pronounced effect than on the large-cap segment.

Another important finding is related to the greater adherence of the Russell 3000 Index to the Russell 1000 Index. This point implies that, from an index composition point of view, an investment in a broad market index is somewhat similar to a large-cap position, and that small caps, then, can be seen as a distinct set, with potential benefits in terms of diversification, as prescribed by Modern Portfolio Theory. To evaluate this aspect, Table 3.1 provides basic descriptive statistics for the indices, with Panel A focusing on daily frequency and Panel B on minute frequency.

In order to test for the correlations among the indices, the test developed by Lawley (1963) is used. In both cases, the Lawley Chi2 test yields p-values of 0.0000. We, therefore, reject the null hypothesis of symmetric correlation and that correlations are not equal to each other. This result confirms the different nature of small caps, as pointed out by the literature, in terms of their risk and return profiles.

*Table 3.1*

**Descriptive Statistics of Russell Indices**

Panel A: Descriptive Statistics – Daily Frequency

| | Russell 3000 | Russell 1000 | Russell 2000 |
|---|---|---|---|
| Initial Date | Oct/18/2021 | Oct/18/2021 | Oct/18/2021 |
| Final Date | Jan/04/2022 | Jan/04/2022 | Jan/04/2022 |
| Sample Size | 54 | 54 | 54 |
| | | | |
| Average Return | 0.0009004 | 0.0009603 | 8.41e-06 |
| Median Return | 0.0015858 | 0.0016299 | 0.000272 |
| Standard Deviation | 0.0091624 | 0.0089614 | 0.014046 |
| Skewness | -0.2264384 | -0.2120141 | -0.1035943 |
| Kurtosis | 3.233298 | 3.2383 | 2.950903 |
| | | | |
| Correlations | | | |
| Russel 3000 | 1.0000 | | |
| Russell 1000 | 0.9987 | 1.0000 | |
| Russell 2000 | 0.8777 | 0.8522 | 1.0000 |

Panel B: Descriptive Statistics – Minute Frequency

| | Russell 3000 | Russell 1000 | Russell 2000 |
|---|---|---|---|
| Initial Date | Oct/19/2021 Open | Oct/19/2021 Open | Oct/19/2021 Open |
| Final Date | Jan/04/2020 Close | Jan/04/2020 Close | Jan/04/2020 Close |
| Sample Size | 20,826 | 20,826 | 20,826 |
| | | | |
| Average Return | 3.38e-08 | 1.22e-07 | -1.99e-06 |
| Median Return | 5.50e-06 | 1.22e-07 | -1.99e-06 |
| Standard Deviation | 0.0002838 | 0.0002861 | 0.0003267 |
| Skewness | -0.1910511 | -0.199024 | 0.1555979 |
| Kurtosis | 11.95432 | 11.9699 | 11.66066 |
| | | | |
| Correlations | | | |
| Russel 3000 | 1.0000 | | |
| Russell 1000 | 0.9922 | 1.0000 | |
| Russell 2000 | 0.7579 | 0.7228 | 1.0000 |

As for the distribution, the minute-wise distribution is remarkably different in terms of the fat-tail situation, in all three cases. This point aligns with the findings of Ahadzie & Jeyasreedharan (2020) related to the direct relationship between the increase in sample frequency and the realized Skewness and Kurtosis. On the other hand, there is not a strong distinction in the asymmetry for Russell 3000 and Russell 1000, as all indices show negative skewness both in minute and daily frequencies. On the Russell 2000, the signal inverts when the frequency is changed, indicating an idiosyncrasy in the high-frequency pattern for the series.

## 3.6. MODEL AND MEASURING DEVICES

In order to understand the specific features of each of the Russell family indices, time series will be subjected to tests under the high-frequency regime. This will allow for comparison among the indices, with potential economic conclusions arising. The research is based on the methodology of realized power variations, as proposed by Ait-Sahalia & Jacod (2012). The model consists of the calculation of test statistics on the realized power variation variable (B), as depicted in Equation (3.4), with three controls being adjusted to isolate the specific components of the time series.

$$B(p, u_n, \Delta n) = \sum_{i=1}^{T/\Delta n} |\Delta_i^n x|^p \, 1_{\{|\Delta_i^n x| \leq u_n\}} \tag{3.4}$$

The first adjustment is related to the power (p), which function is related to the isolation of the continuous or the jump components in the series. When the value of p = 2, the effect of both jumps and continuous components of the series are balanced. This is the result, as explained by Ahadzie & Jeyasreedharan (2020), of the convergence of realized variance to the sample variance in a way that higher frequency sampling may imply more efficient estimates for the variance.

On the other hand, the limits of the third and fourth moments converge to the sum of cubic and quartic jumps, respectively. In this way, an increase in the power variable will highlight the effects of the jumps in the series and disregard the continuous diffusion process.

The second adjustment variable is related to the truncation mechanism ($u_n$), as a form to isolate specific jump sizes. This variable is important in segregating large and small jumps, which allows for the evaluation of its occurrence, in terms of being finite or infinite. At last, the sampling frequency (Δn) is a key element in identifying the limiting behavior of the realized power variation, by comparing the highest available frequency with a larger periodicity.

From the analytical methodology of Ait-Sahalia & Jacod (2012), two test statistics are calculated to identify the presence of Brownian Motion and jump processes in the time series. The first test (sw), used to identify if the series contains the stochastic component, is based on Equation (3.5), where a truncation level is chosen according to a certain number of standard deviations. As the frequency Δn tends to zero, meaning that a 'higher frequency' is selected, the asymptotic behavior of the standard deviation is $\sigma \rightarrow 0$,

and the jump component is eliminated in order to capture the continuous part of the time series. Additionally, the power (p) is set at levels inferior to 2, also to highlight the continuous element.

$$sw(p, u_n, k, \Delta_n) = \frac{B(p,u_n,\Delta_n)}{B(p,u_n,k\Delta_n)} \quad (3.5)$$

In this case, the null hypothesis is defined in the direction of the existence of the Brownian Motion element in the time series, so this defines the first hypothesis of this research. The result of sw converges to 1 in the case where no stochasticity is perceived in the time series and approaches k^(1-p⁄2) when Brownian Motion is present. In this case, the parameter k implies the frequency interpolation element.

The second test statistic (SJ) assesses the presence of jumps in the time series. Equation (3.6) depicts the test, also comparing realized power variations in different frequency patterns mediated by the variable k. Moreover, this test is calculated based on the untruncated form B(p, ∞, Δn) with a power higher than 2. As the test SJ tends to one in its limit, the presence of jumps is detected, while as it tends to zero based on a decay rate of kp/2-1, no jumps are present and the series is based on a continuous Brownian process.

$$s_J(p, \infty, \Delta_n) = \frac{B(p,\infty,k\Delta_n)}{B(p,\infty,\Delta_n)} \quad (3.6)$$

The null hypothesis of the test statistic is related to the presence of jumps in the time series. Accordingly, the second hypothesis of the research states that there will be jumps in the price path of the Russell indices. Moreover, the measurement will focus on potential differences among the indices, that would lead to distinction in the microstructure of them, and therefore differentiate from an inside-out point of view of the volatility dynamics of the series.

## 3.7. EMPIRICAL RESULTS

The empirical result depicts the presence of Brownian Motion on the three Russell indices. For the calculation of SW, the variable power (p) is set between 1 and 1.75 with a 0.25 increment, the truncation varies from 5 to 10 standard deviations under the Δn regime with an increment of 1 standard deviation, the frequency interpolation (k) is set at 2 and 3, and the periodicity Δn is set on 1 and 2 minutes. For each index, then, 96 estimates of sw are calculated and presented in the histograms in Figure 3.2. Such parameters are the same used by Ait-Sahalia & Jacod (2012).

In all cases, the calculation of sw fails to reject the null hypothesis that Brownian Motion is present. Moreover, noise is not a dominating structure as potentially the result of the periodicity (Δn) used. In the case of the Russell 2000 Index, part of the histogram is positioned at a sw below the value of 1, indicating a non-stochastic component in the time series. Evaluating the scatterplot matrix, it is remarkable that the linear relationship between large caps and the broad index is less pronounced for the small caps segment.

*Figure 3.2*

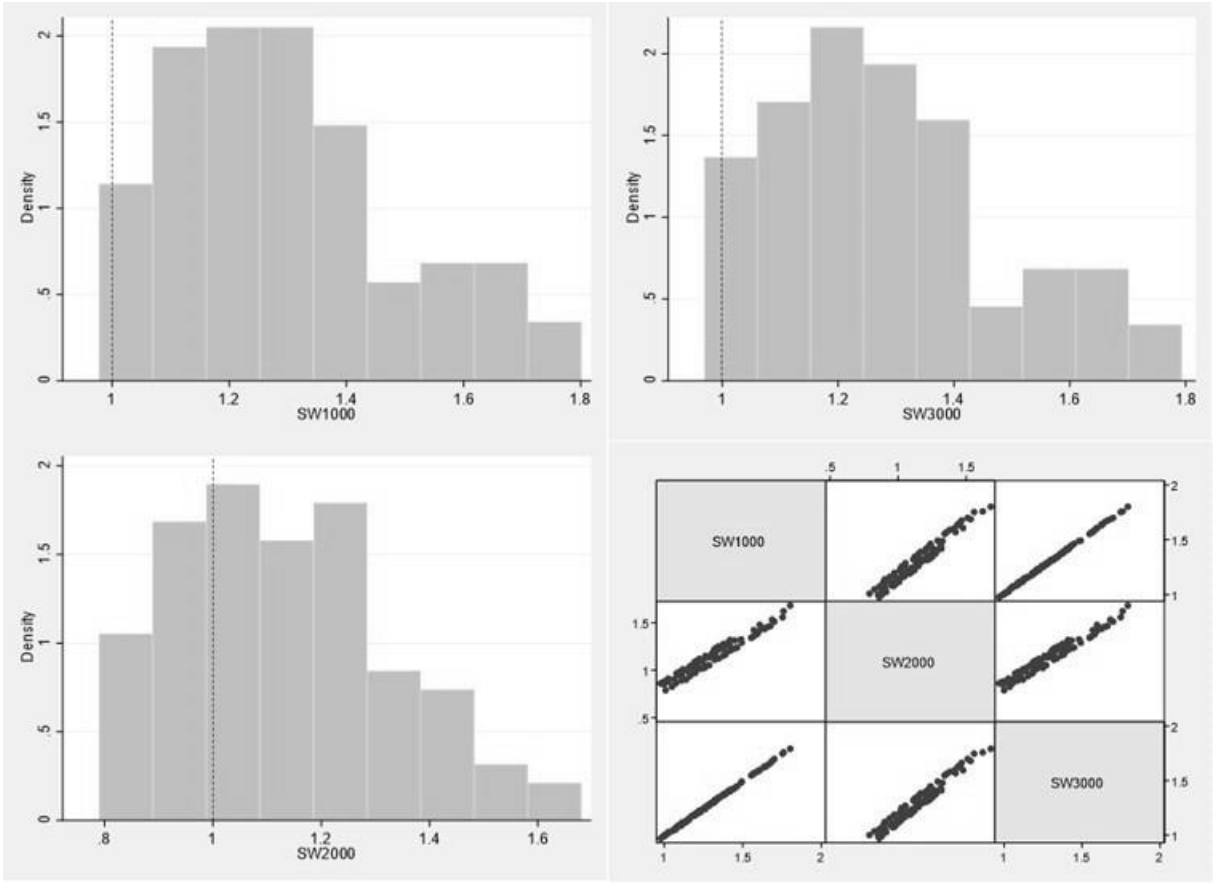
**Histograms for SW (Presence of Brownian Motion)**

Panel A (Top-Left): SW Histogram for Russel 1000 Index

Panel B (Top-Right): SW Histogram for Russel 3000 Index

Panel C (Bottom-Left): SW Histogram for Russel 2000 Index

Panel D (Bottom-Right): Scatterplot Matrix of SW for Russell Indices




Source: Model Calculations

In regards to the presence of jumps, empirical results are calculated using values for the power (p) variable between 3 and 6 with increments of 0.25, frequency interpolation as 2 and 3, and frequency (Δn) defined as 1 and 2 minutes. Under these conditions, a sample of 52 estimates of SJ is calculated for each index.

The null hypothesis that jumps are present is rejected for all three indices, as portrayed in Figure 3.3. This result is potentially the effect of the periodicity of 1 and 2 minutes. In high-frequency works, as in the case of Ait-Sahalia & Jacod (2012), frequencies collected range from 5 seconds to 2 minutes, and therefore,

the data available for this analysis locates on the top-notch of these high-frequency works. Moreover, the authors point out that SJ is found near the value of one in higher frequencies and increases as the periodicity becomes less granular.

*Figure 3.3*

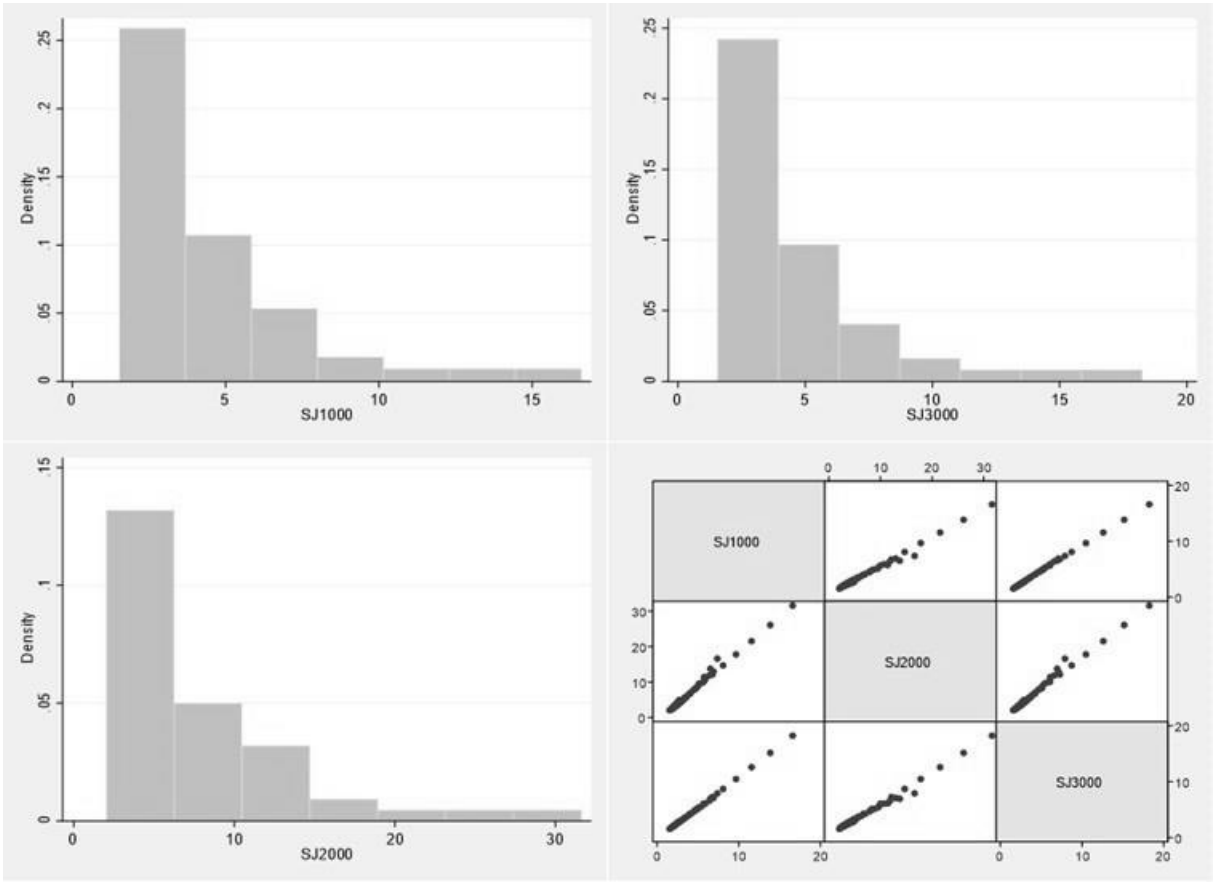
**Histograms for SW (Presence of Jumps)**

Panel A (Top-Left): SJ Histogram for Russel 1000 Index

Panel B (Top-Right): SJ Histogram for Russel 3000 Index

Panel C (Bottom-Left): SJ Histogram for Russel 2000 Index

Panel D (Bottom-Right): Scatterplot Matrix of SJ for Russell Indices




Source: Model Calculations

Most important, on the other hand, is the remarkable difference between the Russell 2000 Index and the other Russell Indices family members. Evidence from Figure 3.3 delineates a longer tail for the Russell 2000 Index in comparison with its large-cap counterpart and the broad market. This finding is relevant in three dimensions. On an operational dimension, the result must be analyzed as presented, in terms of its difference from the other indices. In this case, it implies that the time series behavior of small caps is distinct from large caps, resulting in the de-coupling of risk and return profiles and validating the theoretical landscape of rewarded factors.

On a sample dimension, this result would imply a lower probability of time series jumps for small caps. The point is that this finding may be sample dependent, more remarkably considered in light of smaller returns of small caps in the period. This point is remarkably important as it may imply that small caps may add diversification benefits, when included in portfolios, but not necessarily active return. In other words, excess return that may arise by rewarded factors may be understood as dynamic and transitory.

On the time series functioning dimension, this result imposes, constrained by the sample, two conclusions. The first would be related to the lower probability of jumps in the Russell 2000 Index and the second to a lagged asymptotic trend to jump pattern, as more granular data is introduced. On the other hand, the observed volatility of the Russell 2000 Index is higher than the Russell 1000 or 3000 Indices. In this sense, the research decouples the presence of volatility and jumps, implying that volatility may well arise from the continuous and stochastic element of the time series.

In order to emphasize the distinct pattern of the small caps, an additional evaluation may be significant. The differences in the SJ statistic among the distinct Russell indices are regressed on the power coefficient (p). In this direction, it may be possible to understand how distinct is the microstructure of each segment about each other, and therefore understand potential contrasts in the riskiness and potential of diversification. Table 3.2 depicts these results.

Initially, all models depicted statistical significance for the parameter power and constant. In terms of the power, based on the idea that a higher power coefficient highlights the jump component of the time series, the relevance of the coefficient becomes crucial in defining the relative propensity of a series to jump, in comparison to the other. In this way, this analysis embeds a risk pattern contrast arising from the path disruption in the high-frequency time series.

The evaluation of the three equations highlights the different characteristics of small caps. To comprehend this statement, it may be important to notice that the coefficient calculated for the power coefficient on the difference between large caps and the broad index is -0.1823. In this sense, the test statistics difference, which starts close to a zero level considering that SJ is only applicable at power 3 and above, depicts a higher chance of jumps in the Russell 1000 Index, which grows at a lower degree than the Russell 3000 Index. On the other hand, the degree of separation between those two series is still low implying that, despite the fact that the large-cap is more prone to jumps, the risk pattern of both indices is somewhat equivalent.

In a different direction, the evaluation of the Russell 2000 Index leaves no doubt about a distinct risk pattern for small caps.

*Table 3.2*

**Regression Results in the Difference on Russell Indices**

| | Difference Russell 2000 and 1000 | Difference Russell 1000 and 3000 | Difference Russell 2000 and 3000 |
|---|---|---|---|
| Observations | 52 | 52 | 52 |
| Coefficients and P-Values | | | |
| Power | 2.3124 | -0.1823 | 2.1301 |
| | (0.000) | (0.000) | (0.000) |
| Constant | -7.0963 | 0.5767 | -6.5197 |
| | (0.000) | (0.003) | (0.000) |
| R-Squared | 0.4882 | 0.2930 | 0.5051 |

In the above equations, as much as the difference approaches zero at the lower economic significance of the power, the coefficients of 2.3124 and 2.1301 determine a consistently lower propensity to jump than its peers. As much as this result may appear counterintuitive, it calls for sample dependency. The interesting implication of this point relates to the fact that a position in small caps represents a risk modification of a portfolio, as its riskiness pattern is distinct from large caps, but not necessarily in terms of enlarging expected returns. The success of such investments will depend on the relative momentum of small caps in relation to large caps, which in the last case depends on the allocation skill in terms of timing.

## 3.8. CONCLUSIONS

Empirical results confirm the existence of the specificity of the size factor. As a time series that evolves from a different pattern, in terms of its microstructure, the riskiness of the small-cap segment differentiates itself from the broad market and large caps. As analyzed, this conclusion is not aligned with traditional views that advocate for abnormal returns for this asset class. More specifically, it points to the fact that there is a diversification benefit relative to small caps but the presence of higher-than-market returns will depend on timing. In this sense, the study corroborates with studies and strategies such as Smart Betas.

As an example, active managers may understand the appropriateness of tilting portfolios toward small caps, as active returns may be dependent on market momentum for the asset class and the broad market.

This paper seeks to explore the applications of high-frequency econometrics. In this way, it paves the way for a series of studies that may rely on this sort of data applied to material market questions. For example, additional studies can be done using higher frequency data, exploring other factors, and confirming previous results obtained for the US in international markets. Another important indication for future research is related to the conditions in which jumps occur, such as liquidity or market breadth. At last, a

broader understanding of the extent the absolute returns can be explained by the presence of jumps also may be an important line of research.

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# 4. ARTICLE 3: JUMP IDENTIFICATION AND ITS ENABLERS: AN ULTRA-HIGH-FREQUENCY EMPIRICAL EVALUATION

## 4.1. ARTICLE ABSTRACT

The research evaluates drivers of microstructure risk in the Brazilian stock market in an ultra-high frequency environment. The study confirms the observation of jumps as a general market feature, tracing back to other studies in the US market. Moreover, this research advances the analysis by checking on the homogeneity of the jump feature within different structural subsets of the market, pointing to different microstructural riskiness sources depending on the efficiency level of that specific subset. Specifically, the study depicts a trend towards a noise-driven riskiness for the less efficient market corners while the jump may be a prevalent element for the broad market. As such, the jump may not be understood as a risk enhancer element, prevalent only in specific stocks categorized as "high risk" but rather derived from the scattered nature of information in the market, resulting in price bumps as new and unexpected information is incorporated into prices.

## 4.2. ARTICLE KEY-WORDS

Ultra-high-frequency, jumps, microstructure risk

## 4.3. INTRODUCTION

The introduction of high-frequency econometrics is reshaping the field of equity risk analysis. On a traditional approach, risk has been characterized by exposure to certain factors, especially the market risk factor in which riskiness levels on each stock would be compared to the market portfolio. In such a theoretical landscape, diversification can minimize the portfolio's expected risk based on the combination of compensating risk exposures. One way of thinking about this is reflected in the Modern Portfolio Theory, in which a weak co-movement among assets implies the absorption of risk. This lack of directional adherence by the stocks, in this case, measured by the correlation degree among each pair in the market portfolio, is the result of diversified exposure to the market risk factor.

Despite being a sort of mainstream, many practical problems arise from the simple application of this theory. Fabozzi, Gupta & Markowitz (2002) summarize a key aspect of these issues by recognizing that portfolio optimization, generally based on historical inputs, is not necessarily stable. In this fashion, the inputs used in the optimization suffer from an uncertainty degree, that may be crucial when volatility spikes, as an example.

Without getting into a discussion on normative and descriptive approaches, new forms of measuring and estimating risk have been investigated and developed to improve portfolio performances and risk

management within investment activity. One potential field in which these studies are being developed is high-frequency econometrics, allowed by the unhinged development in information systems in the last years. On one hand, this is the result of the computing capacity expansion and systems interconnectedness, both on exchanges and investor agents. On the other hand, it implies trading strategies that use an extremely short time-lapse. In any case, one of the practical effects of this development is the rise of the necessity to understand microstructure risk.

The link between microstructure risk and traditional econometrics risk is still a point to be better clarified. As explained by Engle (2000), in an environment where time approaches a continuum, information content can be found both in the price of one specific transaction but also in the moment that the transaction occurs. This kind of model would impose a random variable nature to time as well as the content of the time series itself, i.e. price, returns…

To our knowledge, the body of high-frequency studies is still under development in the sense of being able to generate a consistent paradigm for this new econometric field. Moreover, the research up to date tends to focus on the US market, where the degree of information and the average liquidity of assets is broad, therefore paving the way for research feasibility. Although the findings are already promising, one can identify the potential of expanding this research effort to other markets.

From the microstructure risk perspective, a prevalent risk driver can be identified. Initially, risk can be the result of a Brownian Motion, which imposes the characterization of the time series as a normally distributed martingale with an expected mean of zero and variance dependent on time, as explained by Karatzas & Shreve (1998). The second risk source can be associated with the phenomena of noise, in which a mean reversion pattern is based on a stochastic element, as described by Evans (2013), with a direct impact on the observed efficiency of the market, as analyzed by Hasbrouk (1993) and Ait-Sahalia & Yu (2009). At last, the risk process may derive from jumps in the time series, implying actual discontinuation of it as it occurs. According to Ait-Sahalia & Jacod (2012), such innovation in the time series process may be the effect of new information getting into the market or momentary price disruptions, such as a short-period enlargement of bid-ask spreads, for example. The relevant feature, in any way, is related to a real break in the time series continuum, represented by the jump. Goodhart & O'Hara (1997) relate this movement to a break in the enchainment of price, as changing sentiments of sellers and buyers are computed into the price process.

Regarding the discussion about the market, despite the fact that the US market is the source for all prescription financial theories, it shall be depicted much more as a unique case. US capital market development is, in fact, unparalleled in comparison to the rest of the world in terms of breadth, product availability, economic coverage, liquidity, and number (diversity) of actors. In this sense, the replication of theories generated exclusively by the American experience is often conditional to local market realities and, in many cases, a prescription of how the "ideal world" should work rather than a proper explanation of how real markets function.

Moreover, other global markets may present specific insights that can be incorporated into a holistic theory about financial markets. The argument here is that the comparative performance of global markets and the US market may highlight how certain idiosyncrasies play a role in the market, unveiling theoretical features that would stay darkened by the "ideal" conditions of only one case.

This case may be even more important when a new paradigm is under construction, as is the case of high-frequency econometrics. More specifically, the Brazilian stock market (B3) is an interesting source of comparison due to some of its aspects. Firstly, the liquidity in the market is, at the same time, reduced and more unevenly distributed. In this sense, even when the most important equity index (IBOVESPA) is taken into consideration, there will be a relevant disparity in the liquidity features of the components. In such cases, it may allow for an understanding of liquidity as an enabler of time-series disruption. A potential hypothesis in this point resides in the fact that, as liquidity diminishes and the evaluation approaches an ultra-high-frequency landscape, more informational gaps will be presented, tilting the probability of jumps. Another source of explanation would be related to a potential wider bid-ask, as liquidity is reduced, generating conditions for abrupt price movement and therefore breaks on the time series. As these liquidity differences are included in the main stock market index, the comparison among companies can be done with a lower risk of mixing other factors, such as the relative size of the company within the economy.

At last, the Brazilian market has a sound informational transparency level, which allows for data collection and manipulation toward our intended objectives. As such, original econometric studies can be used directly as theory input and result benchmark, as well as our results may be useful for researchers in the global environment.

We depict that the market in general presents a propensity to jumps in its risk microstructure elements. On the other hand, this feature is not homogeneous and noise preponderance is found in less efficient parts of the market.

This research is structured as follows. Session 4.4 presents the theoretical framework in which the jump element can be identified and the test statistics used for this purpose. Session 4.5 depicts the database utilized and comments about the operationalization of the research. Session 4.6 presents the empirical results obtained in terms of jump identification and jump enablers. The research finishes with a conclusion and a description of future research opportunities linked with this one.

## 4.4. THEORETICAL FRAMEWORK

High-frequency econometrics is a new and challenging field with direct implications for finance. On one side, information systems' recent development allowed for the computation of large batches of data, resulting in evident interest in innovative time series specifications in which return trends could be apprehended. On the other hand, exchanges and information systems alike improved data provision services to the market, resulting in just-in-time data availability in a quasi-continuous time fashion.

Such a trend pushed forward the development and management of investments. Nowadays, several commercial models embed intraday calculations, for example. Moreover, trading systems and platforms collect and act on microsecond information, in a mechanism that could relate to arbitrage or according to technical triggers.

This continuous development, nevertheless, has a bottleneck on the theoretical landscape. As time approaches a continuum, traditional econometrics becomes less relevant as there is a change in the observable components of a series. Ait-Sahalia and Jacod (2012) point to the fact that an element that is

a centerpiece in traditional econometrics, such as the drift of a time series, becomes non-observable in high frequency whereas other features like the noise and jumps gain relevance. In a previous study, Ait-Sahalia, Mykland & Zhang (2011) contextualize the noise element as the informational filter of several elements such as trading issues (bid-ask variations, for example), information issues (such as the number of contracts negotiated at a time), and even errors. On another approach, many studies focused on high and ultra-high are concerned with the identification of jumps, that imply actual instant discontinuation of the time series. In this sense, it is plausible to infer that the structural architecture of random variables in a "traditional" time perspective can be traced to a chaotic microstructure nature, consisting of noise, stochasticity, and disruption. Due to this fact, the understanding of this microstructure environment can definitely improve the theoretical understanding of random variables, from a holistic point of view.

Engle (2000) highlights, additionally, that high-frequency possesses a real limit, in which all transactions are recorded as they occur. As a result, the information process on such frequency may not occur at any standardized periodicity, meaning that a time series would potentially present, in this sense, several informational gaps. In this sort of environment, the simple occurrence of a transaction at a specific point in the continuous timeframe embeds informational content, with obvious implications to the understanding of volatility phenomena, for example. Moreover, the author emphasizes that such a framework embeds an important conceptual breakthrough, as time itself can also be considered a random variable.

Based on this finding, using an Autoregressive Conditional Duration (ACD) model, an inverse relationship between the expected duration of the time variable and the volatility of the market variable model is depicted. As such, one could expect volatility spikes linked to a more frequent trading pattern that implies a reduction in the expected time between the trades. Moreover, Ait-Sahalia, Mykland & Zhang (2011) point to a positive relationship between the liquidity of an asset and the existence of a serial correlation of the noise. By combining these two findings, it can be inferred that the serial correlation could be understood as a sort of feedback mechanism for the volatility of an asset, provided that an increase in liquidity can be related to the shorter time duration for the asset. Due to this relationship, the increase in the volatility of an asset may be related, then, to the higher serial correlation of its returns in high frequency.

A central question, in this way, resides in understanding the importance of the jumps, both in terms of its identification but also in terms of its implications related to the volatility of an asset. The reasoning for the point is that, as it refers to a different feature than the noise, there is a potential contradiction in the statements so far: the jump component can be understood as a high volatility momentum, enough to represent a break in the time series pattern, but, at the same time, also potentially reduce serial correlation exactly because it breaks the old pattern. Such a paradigm may suggest that noise and jump imply different volatility components that may be prevalent in specific structural conditions.

In fact, there is no unanimity in the understanding of jumps as a crucial component of the time series. Christensen, Oomen & Podolskij (2014), for example, recognize the importance attributable to the jump component as the literature was developed but recalculate its effective impact on an ultra-high-frequency environment. The authors claim that there is an overstatement of the jump importance as an effect of sampling at a short time interval but yet large enough to keep discrete characteristics. In this view, most of the observed jumps at a sampling frequency of, for example, some seconds, are in reality the result of a "bust of volatility" momentum, without any structural breaks on it. This conclusion leads to a more

discrete role of the jump component on the total variation of an asset, not invalidating, though, the idea that the microstructure of any asset time series is composed of a broader concept of noise.

To elucidate these questions, this research is based on the determination of two hypotheses. The first one states that the jump component is a prevalent element on the market. This hypothesis follows the results of Ait-Sahalia and Jacod (2012), which identified jumps for the Dow Jones in high frequency. Going one step forward, this work tests the continuation of these results in ultra-high frequency. The second hypothesis states that the prevalence of jumps is not homogeneous in the market, implying that different structural characteristics of assets may present distinct micro-structural features.

The methodology used in the research is designed to identify the prevalence of each diffusive or disruptive component within the time series. The authors developed a string of test statistics that are based on the calculation of the Realized Power Variation, as depicted in Equation (4.1).

$$B(p, u_n, \Delta n) = \sum_{i=1}^{T/\Delta n} |\Delta_i^n x|^p \, 1_{\{|\Delta_i^n x| \leq u_n\}} \tag{4.1}$$

Specifically on the theoretical objective of this research, the identification of the jump component may be achieved by a particular specification of parameters in Equation (4.1) to the extent that this effect is highlighted. Ahadzie & Jeyasreedharan (2020) contextualize this specification in terms of Realized Power Variation performance as an unbiased estimator of the actual parameters of a series. According to the authors, the second moment of the time series return is an efficient estimator of the real series variance, when the returns are analyzed under a high or ultra-high frequency, and the number of observations increases. On the other hand, this finding cannot be extrapolated to the third and fourth moments, as predictors of the skewness and kurtosis, for example. In fact, as higher moments are calculated, the greater the departure from its efficient estimation properties as a bias towards the jump component is introduced.

In such a case, Ait-Sahalia and Jacod (2012) consider that a specification of the power parameter *p* in Equation (4.1) equals 2 would then imply balancing both the continuous pattern of the time series and the jumps while the specification of a higher power highlights the jump component and allows of its identification. In this way, when the Realized Power Variation is calculated utilizing a power greater than two, it becomes a proper mechanism to analyze if jumps are present in the sample. On the opposite direction, setting the power variable at a level lower than two would, then, impose a focus on the continuous part of the model.

Another important feature related to the jump identification test statistic resides in the fact that no truncation is executed. Part of the theoretical framework designed is concerned with the determination of small and large jumps. Accordingly, it could be expected that small jumps, when present, may occur finite or infinitely while large jumps may have a finite presence. The mathematical statement follows the economic assumption that small jumps would be related to normal market events such as volatility spikes or bid-ask spread stresses while large jumps should relate to real innovations in the series, like disruptive news entering the market, for example.

Equation (4.2) depicts, then, the test statistic developed for evaluating jumps in the time series. Notice that the resulting model may convey two central and distinct elements: a noise dominated by a stochastic component modeled as a Brownian Motion and a jump element. As previously stated, this characterization stresses the unobservable pattern of the drift in a such small timeframe.

$$s_J(p, \infty, \Delta_n) = \frac{B(p,\infty,k\Delta_n)}{B(p,\infty,\Delta_n)} \quad (4.2)$$

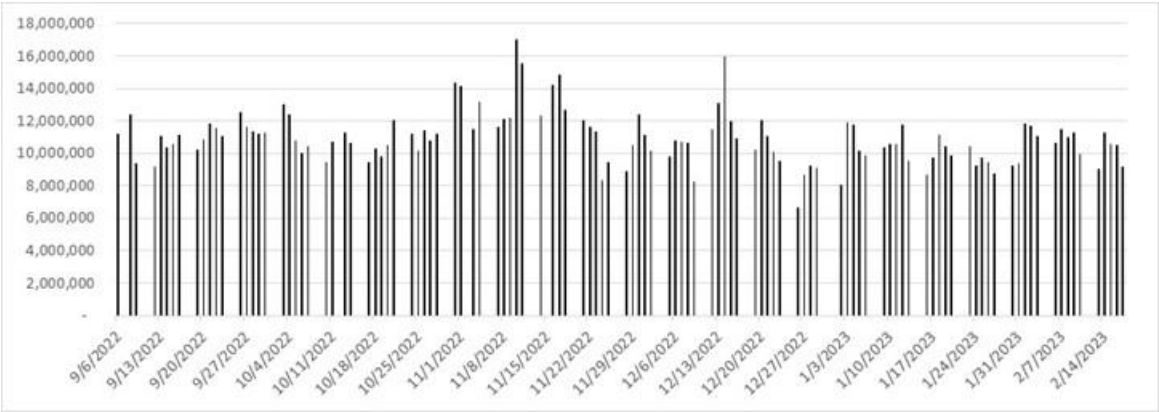
The test statistic $S_J$ will converge to a finite limit when jumps are present in the path of the time series. In this sense, the ratio tends to be 1 and its evaluation can be made graphically on a histogram. On the other hand, time series paths with only continuous stochastic elements will be represented by ratios larger than 1 while a ratio lower than 1 implies the prevalence of white noise and defining components. A last important consideration in the model is that it regards only the above-mentioned three situations (noise, jump, Brownian Motion), meaning that the model is not prepared to capture any sort of pure drift or deterministic behavior, reasonably excluded as such cases do not exist in finance or economics.

## 4.5. DATABASE AND OPERATIONALIZATION

The data that will be used in the research comprises public information from the Brazilian Stock Market (B3), available under the product name 'Market Data'. Each day, B3 releases a file with all recorded transactions for all tickers in the market, using the 1/1000 of a second periodicity. Data collected ranges from the inception of this research phase until the day that the exchange changed the file format, from Sep/06/2022 to Feb/17/2023, representing 114 trading days that include more than 1.2 billion data points. The number of data points per day is depicted in Figure 4.1.

*Figure 4.1*

**Number of Datapoints Recorded Per Day in the Sample**



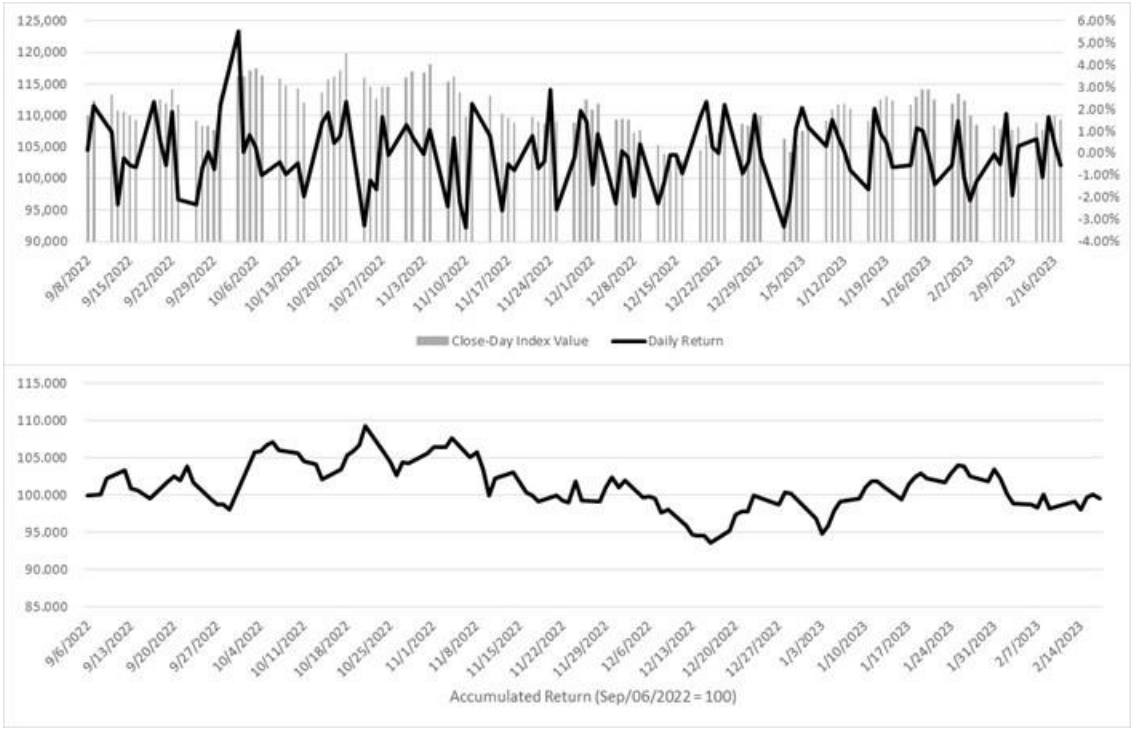
The sample can be considered heterogeneous in terms of trading volumes. If the average day consists of 10.9 million data points, Figure 4.1 depicts volume clusters in November and mid-December and some remarkable low-volume days, especially around the holidays in December.

Figure 4.2 illustrates the time series of the prices and returns of the IBOVESPA index during the sample period. It's remarkable that during the analyzed period, the index had a near zero return despite relevant volatility, with a positive trend during October compensated by a drop in November and a somewhat sideways movement since then. An important consideration when these numbers are evaluated is that most of the sample corresponds to the electoral process for the presidency in Brazil (the first round was held on Oct/2nd and the second round on Oct/30th), which inserts certain idiosyncrasies in stock price movements related to future expectations of government fiscal policy and future economic performance. In addition to this, the sample also captures a major bankruptcy case. In January, Americanas.com, one of the largest retailers in the country, uncovered misconduct in its books, related to accounts payables, that resulted in the revaluation of its debt by a massive amount (BLOOMBERG, 2023). As a result, the equity instruments of the company closed Jan/12th by 2.72 BRL per stock from 12.00 BRL the day before. After the event, and especially while Chapter 11 was under discussion, volatility spikes were noticed in the stock and eventually in some of its peers in the index.

Considering these points, this sample can be considered an interesting and fruitful moment to derive results related to stock market volatility and time series discontinuation.

*Figure 4.2*

**Return of IBOVESPA during the Sample Period**

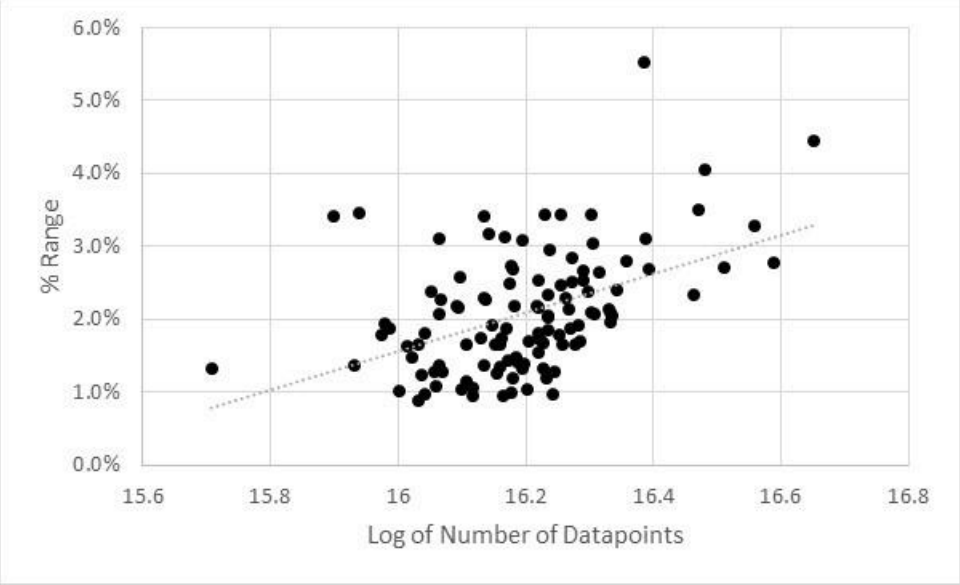


On a general level, the transaction pattern of the stock exchange depicts a strong concentration on Futures contracts on the index and on the US Dollar, around half and one-quarter, respectively, of the total transactions recorded on a specific day. In this way, only a quarter of daily transactions, approximately, will be used on the execution of the research project, subjected yet to additional base cleaning related to other Futures contracts, for example. Moreover, one shall keep in mind that in developing markets like Brazil, the stock exchange presents a sector concentration in financials and infrastructure, creating a relevant breath difference to US markets. An analysis of MSCI (2023, a) and MSCI (2023, b) shows that the Brazilian and American markets have a Herfindahl Index of 0.1722 and 0.1421, respectively, indicating higher concentration in the first. Besides that, the sectors in which the Brazilian market is concentrated (financials, materials, and energy) have lower internal diversification compared to the US (Information Technology, Health Care, and Consumer Discretionary).

This observed concentration may, furthermore, impact directly the general market volatility level. At first glance, an increase in market component homogeneity leads to concentration in the underlying risk factors and potentially to an increase in the correlation within the market. A facet of this volatility profile can be evaluated by the fact that there is a positive and statistically significant relationship between the natural logarithm of the number of data points in ultra-high-frequency in a day and the resulting daily range of prices, measured as the difference between the high and low of the day as a percentage of the opening price of that day. Figure 4.3 reproduces the scatterplot of this relationship. The significance of coefficients and R-Squared of 21.3% are not presented in the chart.

*Figure 4.3*

**Positive Relationship Between the Number of Datapoints and the Daily Range of the Stock**



This result may lead to the conclusion that a higher trading volume leads to an increase in immediate volatility. Litimi, BenSaida & Bouraoui (2016) evaluate such phenomena under the idea of herding-induced volatility and find herding behavior and volatility spikes linked to return movements, volume turnover, and investor sentiment. While the results are not homogeneous for all sectors, it prevails in specific sectors where concentration is noticed for the Brazilian market: basic industries, energy, public utilities, and transportation. Other authors, on the other hand, failed to detect herding in specific markets such as India (Satish & Padmasree, 2018) and Egypt (Mertzanis & Allam, 2018).

Another potential source of volatility could be associated with the increased automated trading under a high-frequency regime. This point may correlate with the above-mentioned herding feature as trading algorithms may share certain characteristics or react to abrupt short-term price movements. Ben Ammar & Hellara (2022) point to a reduction of market volatility associated with high-frequency trading in stable markets. On the other hand, the authors observe spikes in volume, order cancellations, and aggressiveness related to this sort of trading in stressful situations. In this way, automated trading consumes more liquidity than it provides in the market, generating an instantaneous increase in volatility.

Regarding the objectives of this research, we evaluate a potential relationship between the number of transactions and the prevalence of jumps. As jumps are a crucial microstructure element of the riskiness in a time series return path, the idea that volatility spikes should be linked to trading volume shall also be tested for the jump process. Such an analysis is incorporated in Hypothesis 2.

The research will be conducted on the Stata 17.0 BE edition. Operationally, the work was defined around three program codes aiming at importing and formatting data from the stock exchange database, filtering the database on the appropriate time basis, and finally generating the power variance calculation per ticker consolidating them on a complete file. These files may be shared by the authors based on reasonable requests.

The Ibovespa Index includes 89 stocks (B3, 2021). The list of companies and some of the features related to trading securities are available in the appendix. The test statistics are calculated for each ticker for each periodicity (1/1000, 1/100, 1/10, and 1/1 of a second), setting the power parameter from 3 to 6 in increments of 0.25, and sampling frequency was set to 2 and 3. In this way, the histograms reach 9256 SJ observations, which account for a maximum of 15 bins, according to the Sturges Rule.

## 4.6. IDENTIFICATION OF JUMPS AND RESULTS

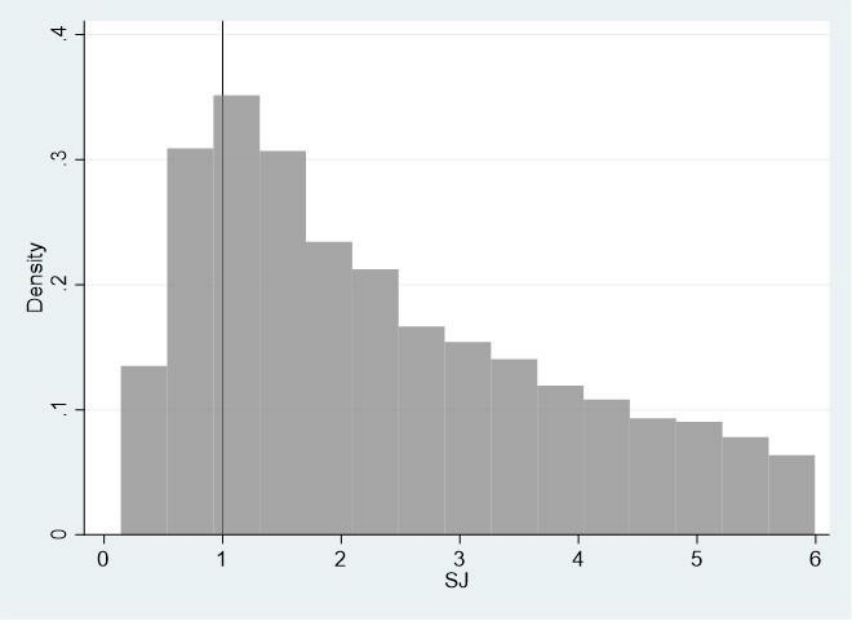
The initial research question to be analyzed is related to the identification of time-series discontinuation to IBOVESPA. The approach utilized regarding this question consists of the evaluation of histogram plots for each of the 89 stocks in the index, unfiltered by any dimensions. The usage of histograms is based on the research conducted by Ait-Sahalia and Jacod (2012). The authors utilize this approach on the results of the test statistic calculated on the Dow Jones Industrial Average based on a quarter periodicity. Moreover, the methodology employed in this research reflects directly on one of the data collection approaches used by the authors, in which all individual components of the Dow Jones Industrial Average are computed.

In this way, the histograms are calculated considering the test statistics designed to identify jumps, SJ considering the number of bins calculated according to the Sturges criteria, for optimal visualization of effects. In the original work, the authors define that jump presence is identifiable when there is a prevalence of results around 1, and a continuous time series pattern when the test result is concentrated on higher value, more specifically higher than 2. Additionally, the authors devise that in cases when the test statistics have greater frequency on a below-one level, the series is dominated by noise. Evaluating the microstructure of the Brazilian index is particularly important as it may indicate if jumps are irrevocably visible in high frequency or if it is a sort of mathematical illusion under some specific sampling frequency, as suggested by Christensen, Oomen & Podolskij (2014). Moreover, the evaluation of an index like Ibovespa is remarkably important as the liquidity difference compared to U.S. markets. This implies another important distinction that may leverage the effect of the ultra-high-frequency sampling and exacerbate the distinctive results in this dataset, compared to the findings of Ait-Sahalia and Jacod (2012).

Regarding this first evaluation, the results indicate the presence of a jump component in the complete time series. In this case, the evaluation of the market's microstructure cannot exclude the existence of timeline breaks. This result is in line with the findings of Ait-Sahalia and Jacod (2012), that identified a jump component for the DJIA index, using the same methodology. From a comparison point of view, it is a remarkable finding that an index with so many differences holds the same results and indicates that jumps may be a common feature of price return time series in high frequency. Linton & Mahmoodzadeh (2018) claim that high-frequency trading may improve, under normal circumstances, price discovery mechanisms, although it may impose systemic risks. The fact that these markets present jump feature do not seem, in this sense, to be a restricting factor to such efficiency gain. Figure 4.4 depicts the total histogram, presented by 9,256 unfiltered observations.

*Figure 4.4*

**Histogram of Sj (Unfiltered Components of Ibovespa)**

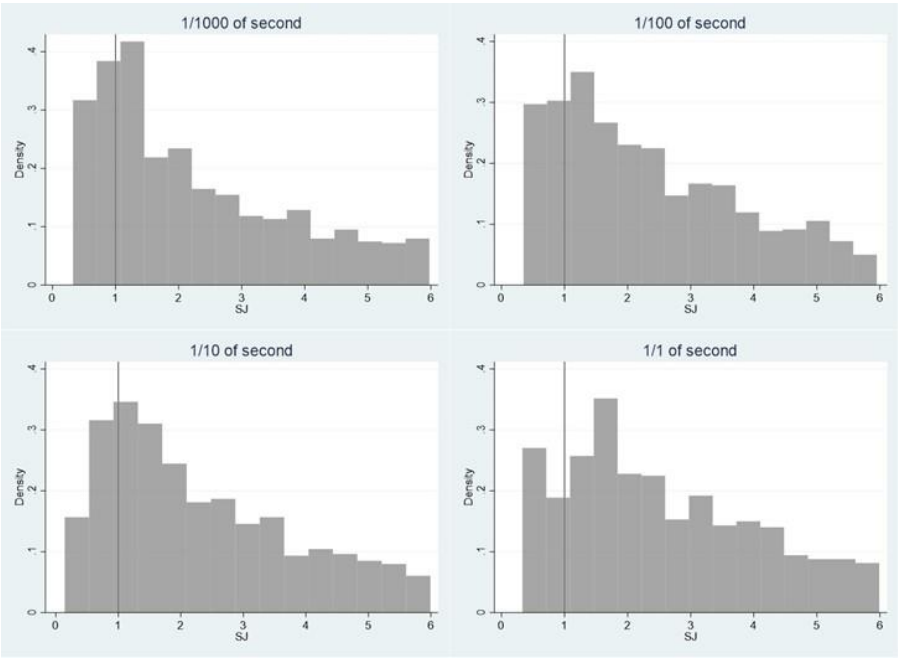


The result presented in Figure 4.4 confirms the first hypothesis of the research, in which jumps are constituent elements of the return time-series path for the market as a whole. Moreover, it may be understood as a confirmation of previous researches that also identify jumps in high-frequency. Going forward with the analysis, the evaluation of the second hypothesis, in this case, will make it possible to understand if this is a homogeneous market feature, independent of idiosyncrasies embedded in certain subsets of the market. In such a way, it gets important to investigate the prevalence of jumps in subsets of the market, leading to a more complete view of the structural features that could play a role in the riskiness of the assets, in comparison to the "normality" case of jump presence depicted by Figure 4.4.

Figure 4.5 depicts the histograms of Sj sorted by the sampling frequency. In general terms, the histograms present frequency spikes in the region that could be defined as jump presence (closer to 1 and not higher than 2). In all cases, the histogram shows lower frequency bins in higher values of $S_j$, indicating that the market cannot be characterized as a continuous stochastic process that could be modeled as a Brownian Motion. In the specific results for 1/100 and 1/1 of a second, the histograms cannot be considered as "well behaved", in terms of depicting a clear top with an up-and-down curve. This point raises important future research questions related to the potential interaction of the jump visualization depending on different sampling periodicity, going from ultra-high-frequency, as analyzed in this research, and high-frequency (something between 1 second and 1 minute). Moreover, the noise component seems also important in explaining the microstructure riskiness, even though it does not seem prevalent in any of the sampling periodicity analyses, resulting in potentially different explanations related to the structure of the market.

*Figure 4.5*

**Histogram of Sj (Sorted by Sampling Frequency)**



The market in Brazil is organized under a specific regulation in which stocks are classified according to their type and governance level. The first feature segregates issues treated as ordinary (ON), a stock that grants voting rights to the holders, and preferential (PN), stocks with advantageous treatment related to claims on cash flows. As can be noticed, the N in the acronyms derives from an old structure where stocks were also classified as nominal, in which the holder is identified, in opposition to a defunct type where the benefits of the issue were carried to the physical holder of the stock.

The second point, related to the governance level, is the result of an effort by the Brazilian exchange to improve the quality of the stocks negotiated in the country, in terms of the norms to assure proper protection to investors. In general terms, this regulation, which aims at developing a healthy stock market, is voluntary from the point of view of the companies listed in the market. As shown in Figure 4.6, most of the companies analyzed in this research are listed in the 'Novo Mercado' (New Market), in which the highest level of governance is required by the company. According to the regulation, minimum requirements to keep the New Market status include the issuance of only ordinary stocks, a minimum of 20% (typically) of free float with a minimum of 10% to the general public, the presence of 20% of independent board members, and stringent financial disclosure and auditing rules, among other requirements (B3, 2023). According to Carrete & Tavarez (2019), firms listed on the 'Nivel 2' (2nd Level) are already required to disclose financial statements using only IFRS or USGAAP and extend some voting rights to preferential stockholders. At 'Nivel 1' (1st Level), an initial requirement related to free-floating is already demanded. Figure 4.6 depicts the frequency of each of these structural features in the index.

At this point, an evaluation of which component is identifiable in the time series can be done in terms of the type of the stock and its governance level. Figure 4.7 depicts the histogram of $S_j$ filtered by the type of stock being transacted. The chart depicts the prevalence of the jump component for ordinary stocks but a preponderance of the noise component for preferential stocks. This result is new and innovative, as it embeds the finding that the price discovery mechanisms for preferential stocks work in line with theoretical assumptions that deviations in fair price should lead to compensatory corrections that would lead to noise in a continuous timeline fashion. Moreover, it marks a departure from the observed prevalent feature in the broad market and the commonly transacted ordinary ones, pointing to a difference in the microstructure riskiness configuration in this specific sector of the market.

On a similar approach, Figure 4.8 depicts the results of $S_j$ when a filter by governance level of stock is applied. In this case, the stocks classified as New Market present a histogram that is aligned with the whole market, whereas lower levels of governance would point to a more hybrid situation (N1) or even noise preponderance (N2). Initially, this result can be seen as counterintuitive in the sense that lower levels of governance tend to present actual continuation regarding its time series, instead of present breaks that could align to potential higher riskiness on these assets.

*Figure 4.6*

**Structural Features of Companies Included in the IBOVESPA**

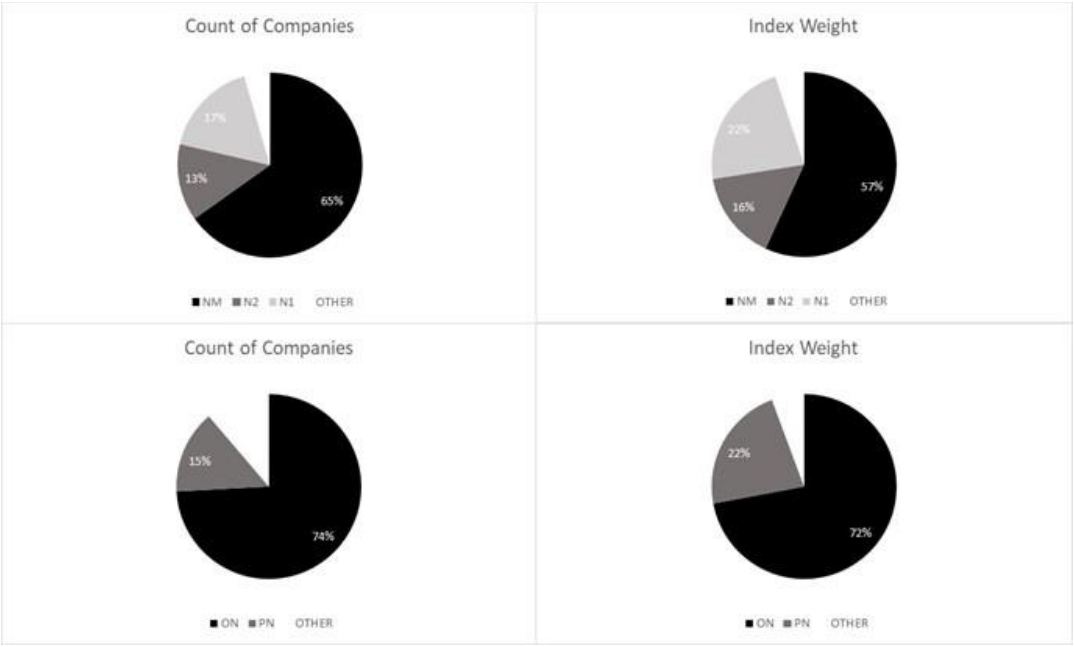



*Figure 4.7*

**Histogram of Sj (Sorted by Type of Stock)**

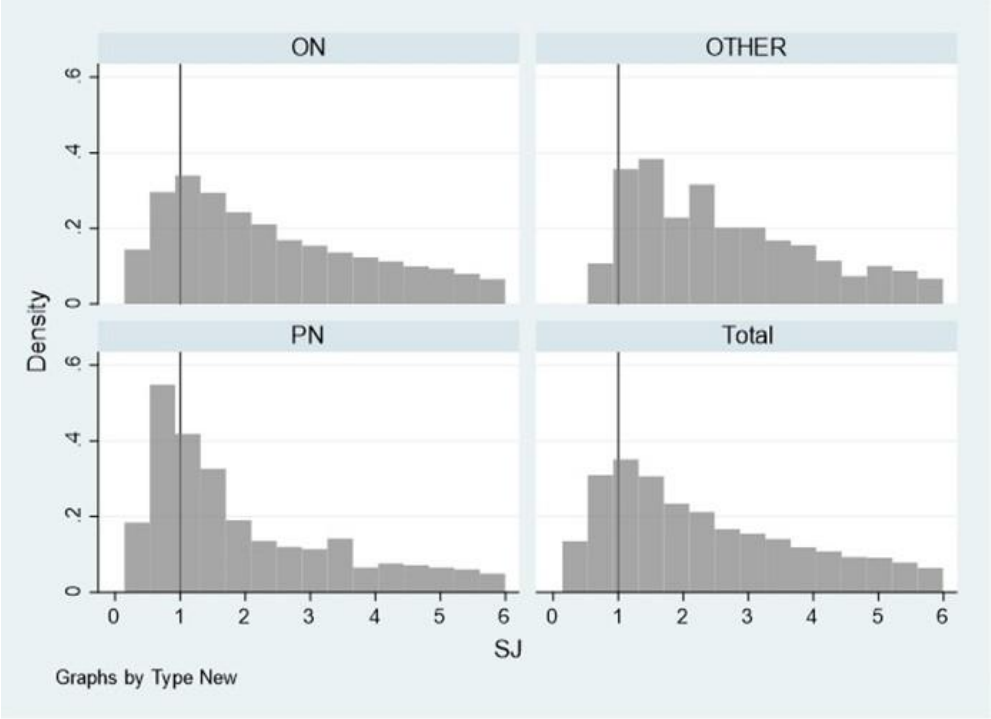
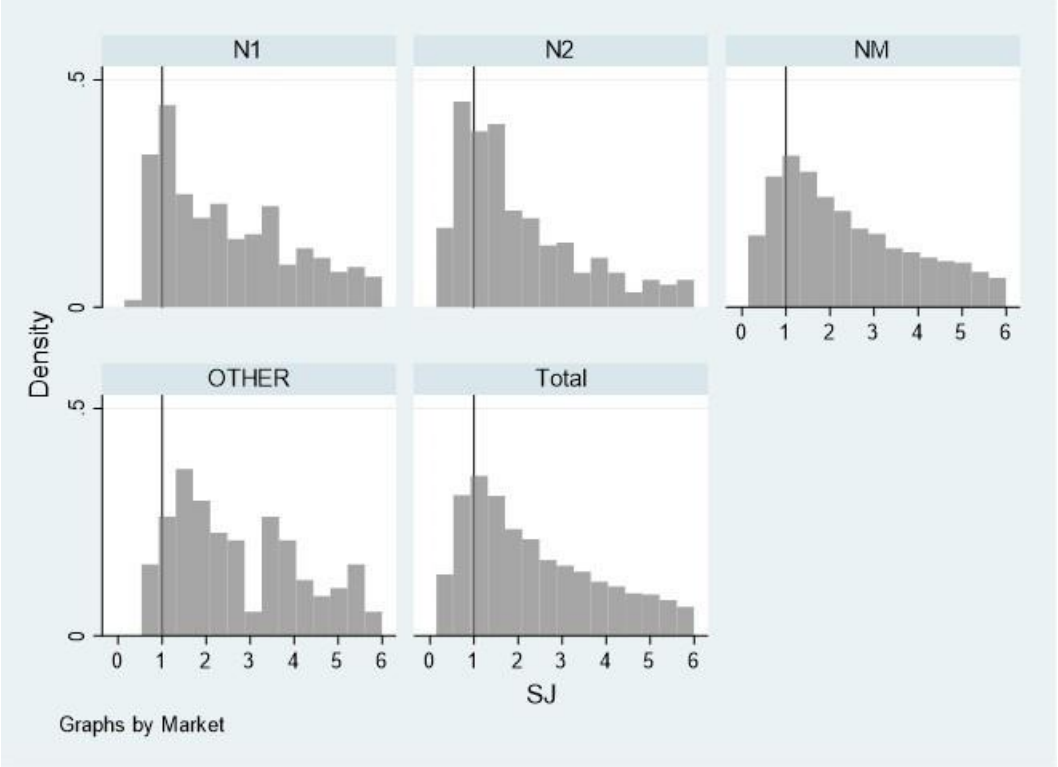


*Figure 4.8*

**Histogram of Sj (Sorted by Governance Level of Stocks)**




On the other hand, the results presented in Figure 4.8 show a departure from what seems to be the normal market feature, the presence of time series breaks. The joint evaluation of Figures 4.7 and 4.8 implies that microstructure riskiness is somewhat different in specific subsets of the market whereas it should be structurally less efficient.

Ait-Sahalia and Jacod (2012) discuss the theoretical limits of their theory in situations when noise is the dominant force in the series. The authors acknowledge that the test statistic is heavily affected by noise, independently of its underlying nature of rounding noise or an additive noise resulting from bounce-backs and directly affected by the series' liquidity or other structural features. Hasbrouk (1993) claims that the standard deviation of the noise is a summary measure of market quality. In the research, the price movement is decomposed into a fundamental random path and the error is measured by the dispersion from the actual price to the first component. In such a way, the greater the occurrence of the noise, the inferior would be the quality of the market. Ait-Sahalia & Yu (2009) observe a relationship between noise and different measures of liquidity. In line with the concepts later explored in Ait-Sahalia, Mykland & Zhang (2011), the authors understand the noise as the practical effect on the price path related to several economic disturbances on bid-ask, discrete price changes, block trade effects, and others. In addition to that, the noise presents a positive correlation to the volatility, both in intraday and monthly measures, spread, cost of trading, and illiquidity ratios.

In this fashion, the results presented to the Brazilian market confirm that noise will be a feature of less efficient subsets of the market. In a broader context, the prevalence of jumps is observed in the market as a whole, as a result of being the expected characteristic of high liquid assets and structurally more efficient market corners. As we move to less efficient structures, such as preferential stocks and lower levels of governance, the higher the propensity to observe noise as a crucial price path driver.

This result, moreover, confirms the stated second hypothesis in the sense that the microstructure riskiness may not be seen as homogeneous but rather dependent on the idiosyncrasy related to the specific subset in terms of efficiency. As liquidity is expected to reflect efficiency, the research also focuses on two analyses for understanding its effect. The first is related to the weight of a specific stock in the total index, as the weight is calculated using a long-term liquidity feature. The second evaluation considers the average number of transactions in specific stocks, a sort of in-sample calculation on liquidity, and therefore, a more short-term approach.

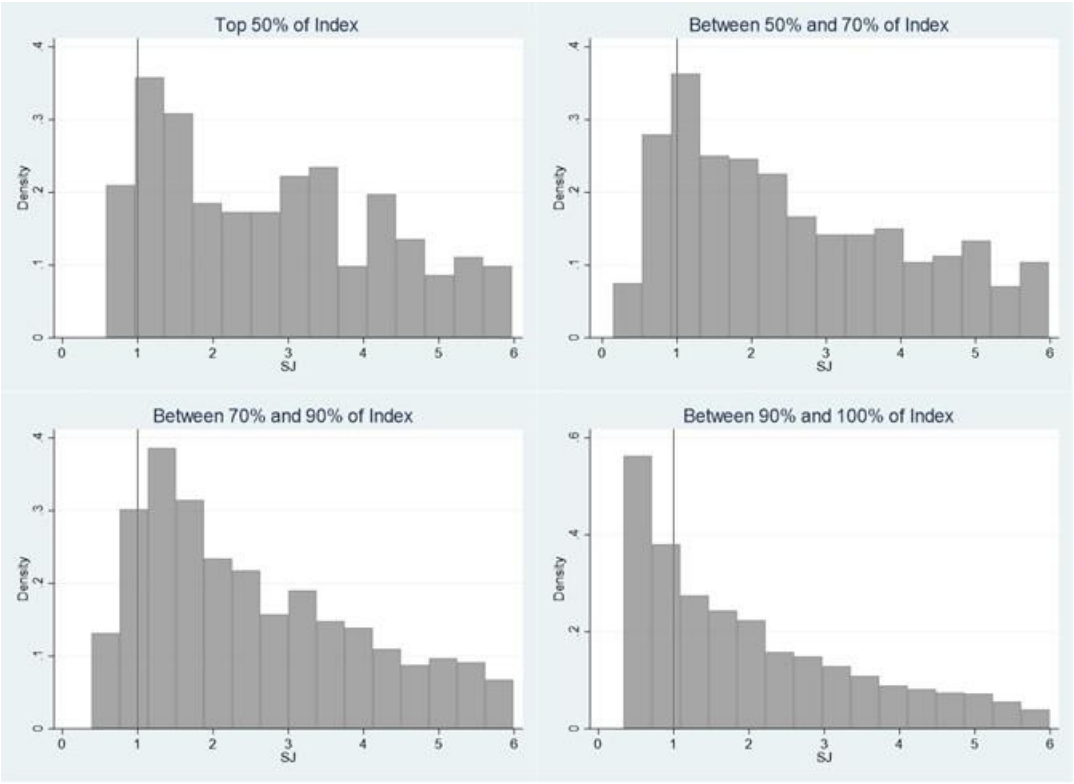
As for the first analysis, a jump component evaluation in regards to the participation of the stock in the market index presents a prevalence of time-series discontinuation for most of the categories while the stocks with the lowest participation in the index are dominated by the noise component. Figure 4.9 depicts the histograms of SJ in four sub-categories representing the weight in the index. The first category represents 50% of the index and includes 9 stocks. The second category consists of an additional 20% in market weight (to a total of 70%) with 12 stocks. The third category groups 27 stocks for another 20% weight in the index (to a total of 90%), and the last category includes the remaining 41 stocks for 10% of the index weight. A list of the stocks may be found in the appendix of the research.

The above-mentioned result confirms the general finding that microstructure riskiness is idiosyncratic in relation to the distinct efficiency levels within different market clusters. Moreover, it points to the presence of a jump component as a sort of regular feature of “normally efficient” market corners.

The second proposed liquidity proxy refers to the average transaction number per day in each stock included in the index. As such, the index was divided into 6 groups of 15 stocks, each one – except the last which counts with only 14 companies, classifying from the group with more transactions to fewer transactions. A list of the stocks may be found in the appendix of the research. Figure 4.10 depicts the histograms for this evaluation.

*Figure 4.9*

**Histogram of Sj (Sorted by Weight in the Index)**




The histograms presented in Figure 4.10 reflect the fact that market clusters are not homogeneous in terms of prevalent microstructure features, with half of the groups presenting the jump component as the most important one while the other half is based on noise. That said, the evaluation of Figure 4.10 does not allow for conclusions related to the expected result that less traded, and therefore less liquid, markets would present lower efficiency levels, captured by the prevalence of the noise characteristic. In fact, groups two, three, and five are the ones in which SJ is concentrated at a level lower than 1.

*Figure 4.10*

**Histogram of Sj (Sorted by Number of Transactions)**

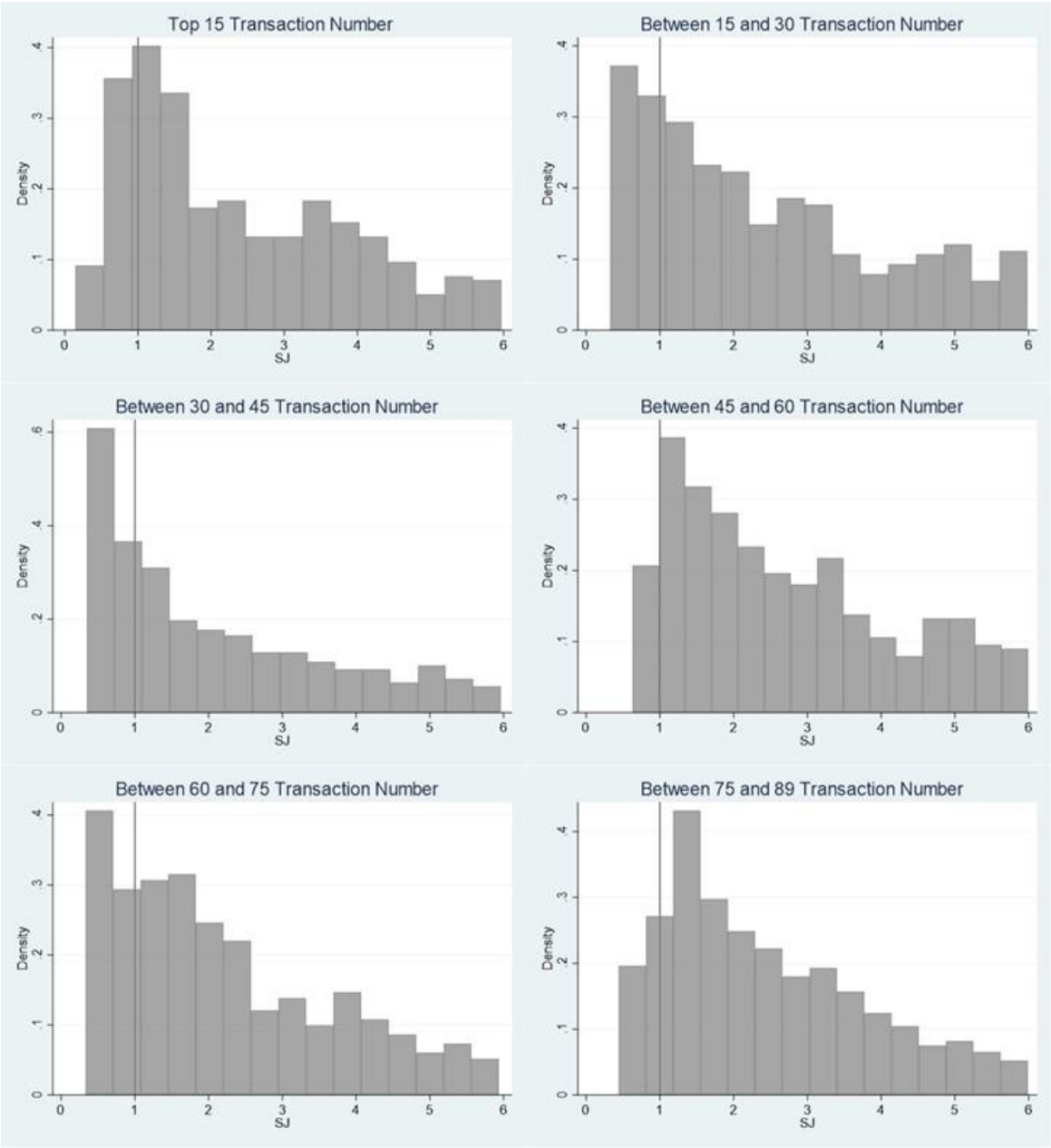
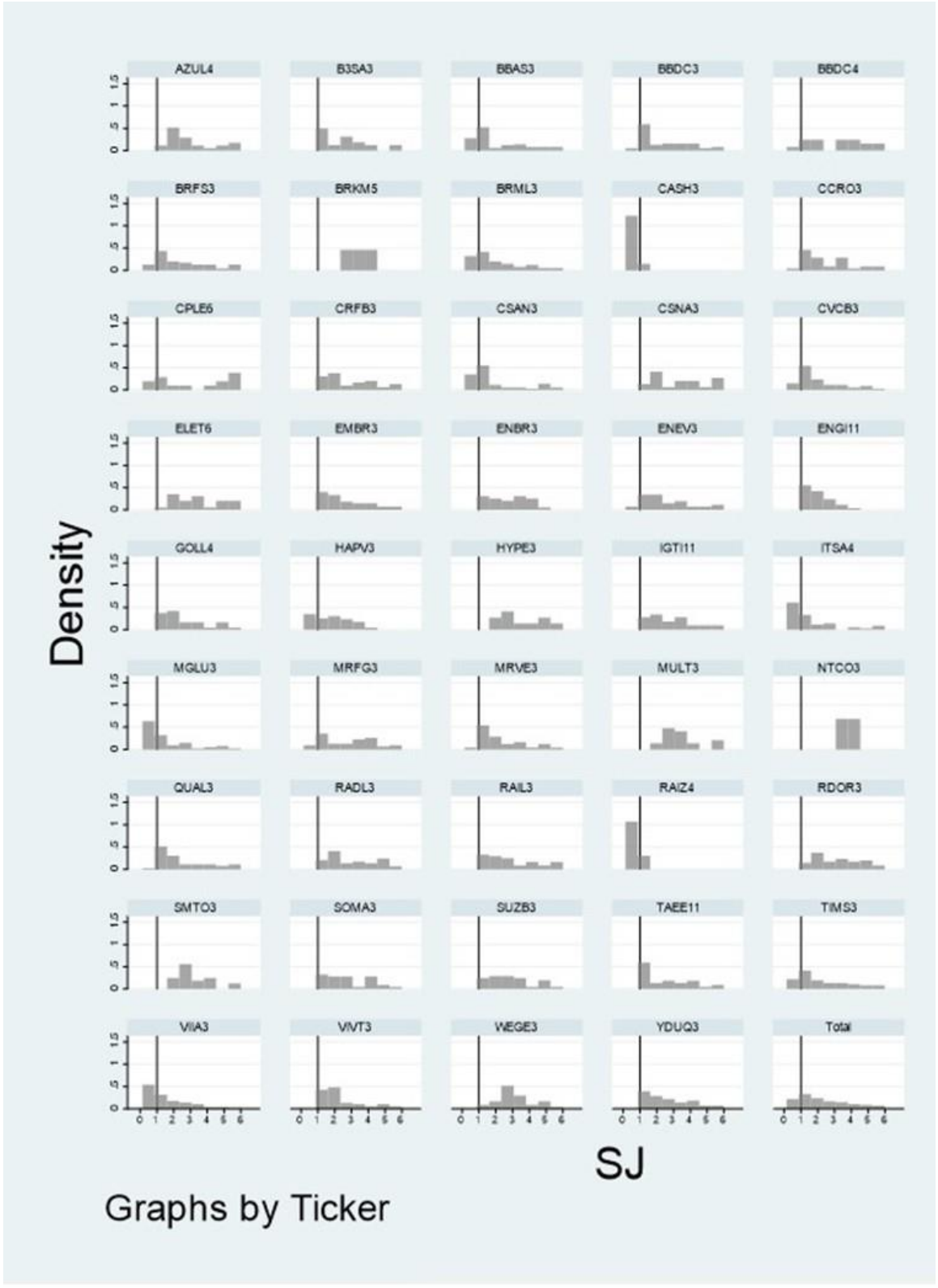


*Figure 4.11*

**Individual Histograms of Companies Included in the IBOVESPA**




AZUL4
B3SA3
BBAS3
BBDC3
BBDC4
BRFS3
BRKM5
BRML3
CASH3
CCRO3
CPLE6
CRFB3
CSAN3
CSNA3
CVCB3
ELET6
EMBR3
ENBR3
ENEV3
ENGI11
GOLL4
HAPV3
HYPE3
IGTI11
ITSA4
MGLU3
MRFG3
MRVE3
MULT3
NTCO3
QUAL3
RADL3
RAIL3
RAIZ4
RDOR3
SMTO3
SOMA3
SUZB3
TAEE11
TIMS3
VIIA3
VIVT3
WEGE3
YDUQ3
Total
Density
SJ
Graphs by Ticker

This result may suggest some conclusions. One of them can be related to the potential fact that short-term liquidity measures may not be ideal to evaluate market efficiency while long-term measures may impose a certain stability for a trend to be captured. A second potential conclusion is that, in the short-term, number of transactions may spike for almost any stock with strong price movement, for example, while this is a transitory situation that cannot reflect the regular riskiness of that stock. To further investigate such ideas, Figure 4.11 depicts the individual histograms for all stocks considered. The evaluation of Figure 4.11 may confirm if there is any other missing condition that should be evaluated.

The individual evaluation of Figure 4.11 confirms a heterogeneous stock base. The focus on the stocks in which the time series process is dominated by noise implies some interesting empirical remarks. There are nine stocks with clear trends pointing to noise prevalence. Before getting into the stocks, it is important to raise a cautionary note. The above-mentioned subsets generally count with more stocks, and the combined result of these subsets is not only attributable to the presence of any of these 9 stocks but rather the combination of all stocks within the cluster (including the fact that the cluster analysis is also benefited by a broader sample). In this way, the evaluation of Figure 4.11 may be seen as complementary to the previous conclusions in this research.

Of the nine stocks, three are not related to ordinary stocks. Among them, Itausa (ITSA4) refers to the investment holding that keeps some relevant participation in other companies (ITAUSA, 2023) including a 37% stake in Itau (ITUB4), 38% in Dexco (DXCO3), a 30% participation in Alpargatas (ALPA4) and a 10% in CCR (CCRO3). As a result, one can understand Itausa as a diversified company, with interests raging in several economic sectors like financials (ITUB4), materials (DEXCO3), consumer discretionary (ALPA4), and infrastructure (CCRO3), among others, which could lead to a characteristic of efficiency. Moreover, all stocks included in the ITSA4 business umbrella and evaluated in this research present a prevalent feature of a jump in ultra-high frequency, which could be also expected to carry over to ITSA4. On the other hand, ITSA4 is still dominated by a noise-driven process, leading to a conclusion that microstructure riskiness may result from investor activity, rather than company fundamentals.

The other two companies are Raizen (RAIZ4), a sugar-cane-based fuel company, and Usiminas (USIM5), a steelworks company. In both cases, they account for small participation in the index, around 0.2% each, combined with a non-ordinary trading vehicle. Remarkably, on the other hand, this combination alone did not produce the same effects on AZUL4, ALPA4, BPAN4, and GOLL4, when these stocks are evaluated individually. In this same direction, four out of six ordinary stocks seem to be more thinly traded and count with low weight in the index: Cielo (CIEL3), a payment processing company, Cogna (COGN3), an education company, Eco Rodovias (ECOR3), an infrastructure company, and Meluiz (CASH3), another payment processing company. Alone, these features cannot explain the noise prevalence.

A last interesting empirical consideration, regarding the noisy stocks, can be made in the cases of Magazine Luiza (MGLU3) and Via Varejo (VIIA3). Both companies operate in the retail sector, which was affected by the failure of Americanas (AMER3) in January, sparking a spill-over effect that hit both stocks. As an effect, despite presenting small weight in the index, 0.405% and 0.195% respectively, both stocks recorded spikes in trading activity. MGLU3 and VIIA3 had the 6th and 23rd positions in the average transaction number. A similar movement is found in AMER3, the 15th most traded stock in the period with only 0.303% of the index. Surprisingly, while both MGLU3 and VIIA3 presented a noise-dominated time series path, AMER3, which can be seen as the source of the market disruption, is presented as a jump-dominated stock. Empirically, it is rational that AMER3 may be dominated by a jump component as

the stock had very aggressive price movements. On the other hand, this case may be distinct from a general market jump conclusion. Please refer to the Future Research session for more comments on this topic.

A sort of surprising result is related to the fact that eleven stocks are driven purely by stochastic continuous movements not related to noise or jumps. These companies are Localiza (RENT3), a car rental company, Petrobras (PETR3), the state-owned oil giant, 3R (RRRP3), a private oil and gas company, Santander (SANB11), the Brazilian operation of the Spanish bank, Gerdau (GGBR4), a steelworks company, Lojas Renner (LREN3), a clothing and apparel retailer, Eletrobras (ELET3), an utility, Braskem (BRKM5), an energy producer, Weg (WEGE3), a heavy equipment manufacturer, Multiplan (MULT3), a shopping mall manager, and Natura (NTCO3), a beauty products producer. Considering the histograms presented in Figure 4.11, these stocks can be characterized as a result of pure Brownian Motion processes. As a consequence, it could be expected that the resulting riskiness level would be moderated by this fact. The empirical conclusion, on the other hand, does not support this expectation. Among the stocks in the group, the observed volatility is not generally among the lowest in the index, considering both coefficient of variation or price range.

*Figure 4.12*

**Kurtosis and Skewness Scatterplot with Selected Stocks Highlighted (1/1000 Second)**

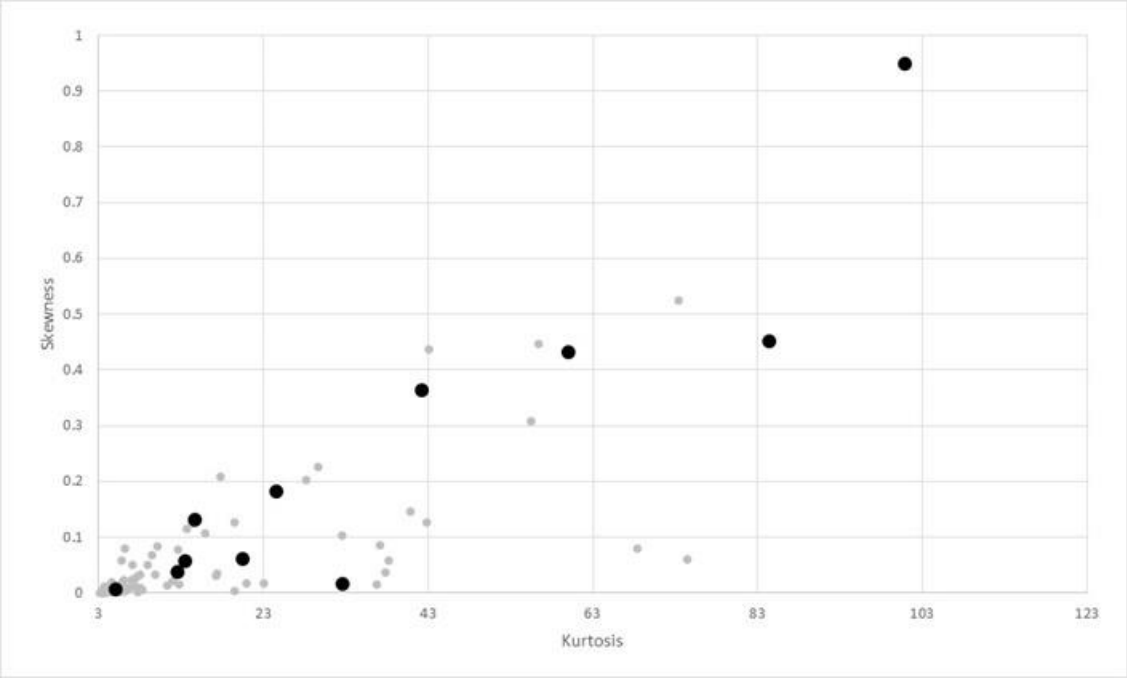



Figure 4.12 depicts the observed skewness and kurtosis of the stocks in the index, with this group of stocks highlighted in black. In the figure, it is possible to evaluate the degree to which each stock departs from a normal distribution, in a 1/1000 second periodicity.

The evaluation of Figure 4.12 depicts that the selected stocks present unexpectedly high observed kurtosis and skewness, indicating that they tend to depart from a normal distribution. This result is specifically trilling as the Brownian Motion, which is expected to be the primary process driver for this kind of stock is specifically modeled based on a normal distribution assumption. Further studies may evaluate this point in the future.

## 4.7. CONCLUSION

This study confirms the observation of jumps as a general market feature, in line with previous findings in high-frequency econometrics, especially the one obtained by Ait-Sahalia and Jacod (2012). Such a result is relevant as it points to a common output for two very different markets and indices, the Ibovespa in Brazil and the Dow Jones in the US. Moreover, this research advances the analysis by checking on the homogeneity of the jump feature within different structural subsets of the market. The results point to different microstructural riskiness sources depending on the efficiency level of that specific subset, as discussed by Hasbrouk (1993). Specifically, the study depicts a trend towards a noise-driven riskiness for the less efficient market corners while the jump may be a prevalent element for the broad market.

In this sense, the jump may not be understood as a risk enhancer element, prevalent only in specific stocks categorized as “high risk”. In reality, the jump is derived from the scattered nature of information in the market, resulting in price bumps as new and unexpected information is incorporated into prices. This movement is, therefore, a natural part of the volatility profile. On the other hand, the jump process may also be the result of a disruptive situation, such as the one observed for AMER3. In such cases, the jump is also a result of risk enhancer news. Evaluating the size of jumps, as suggested by Ait-Sahalia and Jacod (2012), may be key to differentiating if a jump results from a risk enhancement process.

## 4.8. FUTURE RESEARCH

The understanding of high-frequency and ultra-high-frequency econometrics has room for many new studies. This research indicates some important gaps that can be further explored to improve the knowledge base in this field.

Initially, the definition of the test statistics is done on a somewhat cross-section basis. A theoretical landscape could allow for the evaluation of a changing propensity of a jump over time. Such an analysis would improve the understanding of the relationship between volume and jumps or noise, for example.

Further research may also be conducted in understanding the stocks that, on a stand-alone basis, derive their riskiness from a pure Brownian Motion movement but fail to comply with the probability distribution that governs it. This paradigm may uncover important findings on the riskiness of equities.

Another important point refers to further testing for the serial correlation of errors, regarding the liquidity of the asset and the effect on the total noise recorded. This evaluation could lead to a more precise specification of environmental conditions for time-series breaks.

At last, an empirical analysis of companies under severe stress may improve the understanding of tail risk situations. This research briefly touched on the AMER3 case. A case study on this company and a comparison with other previous cases may be very enlighting.

## 4.9. APPENDIX

*Table 4.1*

**Appendix – List of Stocks (IBOV - Portfolio as of Jan. to Apr. 2023)**

| Ticker | Company Name | Type | Market | Part (%) | Cat. Part (%) | Number of Transactions | Cat. Transactions |
|---|---|---|---|---|---|---|---|
| VALE3 | VALE | ON | NM | 15.51 | 1 | 9,016,248 | 1 |
| ITUB4 | ITAUUNIBANCO | PN | N1 | 6.17 | 1 | 7,326,416 | 1 |
| PETR4 | PETROBRAS | PN | N2 | 5.75 | 1 | 13,004,575 | 1 |
| PETR3 | PETROBRAS | ON | N2 | 5.02 | 1 | 3,857,195 | 1 |
| ELET3 | ELETROBRAS | ON | N1 | 4.05 | 1 | 4,093,145 | 1 |
| BBDC4 | BRADESCO | PN | N1 | 4.02 | 1 | 6,704,018 | 1 |
| B3SA3 | B3 | ON | NM | 3.99 | 1 | 6,355,616 | 1 |
| ABEV3 | AMBEV S/A | ON |  | 3.28 | 1 | 4,476,100 | 1 |
| WEGE3 | WEG | ON | NM | 2.61 | 1 | 2,834,726 | 2 |
| BBAS3 | BRASIL | ON | NM | 2.54 | 2 | 5,452,204 | 1 |
| ITSA4 | ITAUSA | PN | N1 | 2.29 | 2 | 3,419,376 | 2 |
| RENT3 | LOCALIZA | ON | NM | 2.12 | 2 | 3,778,959 | 1 |
| SUZB3 | SUZANO S.A. | ON | NM | 1.80 | 2 | 2,711,983 | 2 |
| RDOR3 | REDE D OR | ON | NM | 1.75 | 2 | 2,096,635 | 4 |
| GGBR4 | GERDAU | PN | N1 | 1.65 | 2 | 3,387,299 | 2 |
| PRIO3 | PETRORIO | ON | NM | 1.61 | 2 | 3,629,736 | 2 |
| EQTL3 | EQUATORIAL | ON | NM | 1.53 | 2 | 2,452,258 | 3 |
| BPAC11 | BTGP BANCO | UNT | N2 | 1.51 | 2 | 3,781,474 | 1 |
| RADL3 | RAIADROGASIL | ON | NM | 1.46 | 2 | 2,327,783 | 3 |
| JBSS3 | JBS | ON | NM | 1.28 | 2 | 2,462,026 | 3 |

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|---|---|---|---|---|---|---|---|
| BBSE3 | BBSEGURIDADE | ON | NM | 1.16 | 2 | 2,623,325 | 2 |
| HAPV3 | HAPVIDA | ON | NM | 1.16 | 3 | 5,221,322 | 1 |
| RAIL3 | RUMO S.A. | ON | NM | 1.16 | 3 | 2,851,838 | 2 |
| BBDC3 | BRADESCO | ON | N1 | 1.05 | 3 | 1,471,575 | 5 |
| CSAN3 | COSAN | ON | NM | 1.03 | 3 | 2,876,404 | 2 |
| LREN3 | LOJAS RENNER | ON | NM | 1.02 | 3 | 4,506,420 | 1 |
| SBSP3 | SABESP | ON | NM | 1.00 | 3 | 2,142,144 | 3 |
| ENEV3 | ENEVA | ON | NM | 0.97 | 3 | 2,620,148 | 2 |
| ASAI3 | ASSAI | ON | NM | 0.94 | 3 | 3,122,644 | 2 |
| HYPE3 | HYPERA | ON | NM | 0.93 | 3 | 2,047,096 | 4 |
| VBBR3 | VIBRA | ON | NM | 0.89 | 3 | 2,661,779 | 2 |
| CMIG4 | CEMIG | PN | N1 | 0.83 | 3 | 2,210,295 | 3 |
| VIVT3 | TELEF BRASIL | ON | ON | 0.81 | 3 | 1,390,570 | 5 |
| TOTS3 | TOTVS | ON | NM | 0.74 | 3 | 2,516,737 | 3 |
| KLBN11 | KLABIN S/A | UNT | N2 | 0.72 | 3 | 2,149,707 | 3 |
| UGPA3 | ULTRAPAR | ON | NM | 0.71 | 3 | 1,987,888 | 4 |
| CPLE6 | COPEL | PNB | N2 | 0.64 | 3 | 2,040,972 | 4 |
| ELET6 | ELETROBRAS | PNB | N1 | 0.61 | 3 | 1,464,130 | 5 |
| ENGI11 | ENERGISA | UNT | N2 | 0.57 | 3 | 1,271,100 | 6 |
| CCRO3 | CCR SA | ON | NM | 0.56 | 3 | 2,040,263 | 4 |
| EMBR3 | EMBRAER | ON | NM | 0.54 | 3 | 1,521,480 | 5 |
| SANB11 | SANTANDER BR | UNT | | 0.52 | 3 | 1,158,716 | 6 |
| TIMS3 | TIM | ON | NM | 0.52 | 3 | 1,866,210 | 4 |
| NTCO3 | GRUPO NATURA | ON | NM | 0.50 | 3 | 3,429,827 | 2 |
| EGIE3 | ENGIE BRASIL | ON | NM | 0.50 | 3 | 961,717 | 6 |
| BRFS3 | BRF SA | ON | NM | 0.46 | 3 | 2,693,021 | 2 |
| GOAU4 | GERDAU MET | PN | N1 | 0.45 | 3 | 1,802,907 | 5 |
| CSNA3 | SID NACIONAL | ON | EJ | 0.44 | 3 | 1,835,576 | 4 |

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|---|---|---|---|---|---|---|---|
| MGLU3 | MAGAZ LUIZA | ON | NM | 0.41 | 4 | 5,699,096 | 1 |
| CRFB3 | CARREFOUR BR | ON | NM | 0.40 | 4 | 1,702,044 | 5 |
| TAEE11 | TAESA | UNT | N2 | 0.39 | 4 | 1,379,175 | 5 |
| BRAP4 | BRADESPAR | PN | N1 | 0.38 | 4 | 1,279,668 | 5 |
| RRRP3 | 3R PETROLEUM | ON | NM | 0.38 | 4 | 2,048,116 | 4 |
| BRML3 | BR MALLS PAR | ON | NM | 0.35 | 4 | 1,561,428 | 5 |
| BRKM5 | BRASKEM | PNA | N1 | 0.32 | 4 | 1,554,400 | 5 |
| CPFE3 | CPFL ENERGIA | ON | NM | 0.32 | 4 | 1,085,299 | 6 |
| AMER3 | AMERICANAS | ON | NM | 0.30 | 4 | 3,730,604 | 1 |
| MULT3 | MULTIPLAN | ON | N2 | 0.30 | 4 | 2,193,027 | 3 |
| CIEL3 | CIELO | ON | NM | 0.30 | 4 | 2,899,782 | 2 |
| FLRY3 | FLEURY | ON | NM | 0.28 | 4 | 1,204,629 | 6 |
| ENBR3 | ENERGIAS BR | ON | NM | 0.26 | 4 | 1,267,846 | 6 |
| SOMA3 | GRUPO SOMA | ON | NM | 0.26 | 4 | 2,141,280 | 3 |
| ARZZ3 | AREZZO CO | ON | NM | 0.24 | 4 | 810,558 | 6 |
| CMIN3 | CSNMINERACAO | ON | N2 | 0.23 | 4 | 1,068,401 | 6 |
| SLCE3 | SLC AGRICOLA | ON | NM | 0.23 | 4 | 836,557 | 6 |
| RAIZ4 | RAIZEN | PN | N2 | 0.23 | 4 | 2,293,192 | 3 |
| IGTI11 | IGUATEMI S.A | UNT | N1 | 0.20 | 4 | 1,936,951 | 4 |
| COGN3 | COGNA ON | ON | NM | 0.20 | 4 | 2,539,523 | 3 |
| SMTO3 | SAO MARTINHO | ON | NM | 0.20 | 4 | 1,230,422 | 6 |
| VIIA3 | VIA | ON | NM | 0.20 | 4 | 2,871,402 | 2 |
| USIM5 | USIMINAS | PNA | N1 | 0.19 | 4 | 1,982,816 | 4 |
| AZUL4 | AZUL | PN | N2 | 0.19 | 4 | 1,923,793 | 4 |
| CYRE3 | CYRELA REALT | ON | NM | 0.18 | 4 | 2,460,603 | 3 |
| BEEF3 | MINERVA | ON | NM | 0.17 | 4 | 2,176,297 | 3 |
| ALPA4 | ALPARGATAS | PN | N1 | 0.16 | 4 | 1,803,658 | 4 |
| YDUQ3 | YDUQS PART | ON | NM | 0.16 | 4 | 1,931,926 | 4 |

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|---|---|---|---|---|---|---|---|
| LWSA3 | LOCAWEB | ON | NM | 0.15 | 4 | 1,818,299 | 4 |
| MRFG3 | MARFRIG | ON | NM | 0.14 | 4 | 2,138,784 | 3 |
| PCAR3 | P.ACUCAR-CBD | ON | NM | 0.13 | 4 | 1,321,463 | 5 |
| MRVE3 | MRV | ON | NM | 0.12 | 4 | 2,488,305 | 3 |
| PETZ3 | PETZ | ON | NM | 0.11 | 4 | 2,028,947 | 4 |
| BPAN4 | BANCO PAN | PN | N1 | 0.11 | 4 | 868,748 | 6 |
| DXCO3 | DEXCO | ON | NM | 0.10 | 4 | 1,371,557 | 5 |
| QUAL3 | QUALICORP | ON | NM | 0.08 | 4 | 1,195,854 | 6 |
| GOLL4 | GOL | PN | N2 | 0.07 | 4 | 1,344,465 | 5 |
| ECOR3 | ECORODOVIAS | ON | NM | 0.07 | 4 | 1,264,576 | 6 |
| EZTC3 | EZTEC | ON | NM | 0.07 | 4 | 1,096,385 | 6 |
| CVCB3 | CVC BRASIL | ON | NM | 0.06 | 4 | 1,383,665 | 5 |
| CASH3 | MELIUZ | ON | NM | 0.04 | 4 | 1,367,045 | 5 |

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# 6. MATHEMATICAL APPENDIX

Log-Linear Model (DeFusco et al., 2007)

$$y_t = e^{b_0 + b_1 t}$$

Trend is result of a continuous compounding with ($e^t$-1) as constant growth rate. As both sides are met on logarithm, it results in the log-linear function on Equation 2.

ARCH Model (Engle, 1982)

Starting with a model where current error is dependent on past error $y_t = \varepsilon_t h^{1/2}_{t-1,}$ that makes conditional variance equals to $h_t = \alpha_0 + \alpha_1 y^2_{t-1.}$ In this way, $\varepsilon_t \approx N(0,h_t)$ where α is a vector of unknown parameters.

Mean Reversion Pattern in Random Walks

In AR (1) model, $x_t = b_v + b_1 x_{t-1}$ meaning that the mean reversion pattern is equal to $x_t = \frac{b_0}{(1-b_1)}$. In the case of a 'Random Walk', the coefficients of the series will assume the level of $b_0$=0 and $b_1$=1, what will generate a mean reversion to 0/(1-1), what is not defined.

Variance Proportional to Time in Random Walks

In a 'Random Walk' where $x_1$=0, then $x_2$=0+$\varepsilon_{2,}$ so the variance of the error in $x_2$ is equal to $Var(\varepsilon_2)=\sigma^2$. As another time point is introduced, the model becomes $x_3=x_{2+}\ \varepsilon_{3,}$ substituting for previous $x_2$ equation, $x_3=\varepsilon_{2+}$ $\varepsilon_3$, that are both random and independent errors. In this way, the variance of the error is $E(\sigma^2)=Var(\varepsilon_2)+Var(\varepsilon_3)=2\sigma^2=(t-1)\sigma^2$

First Degree Differentiation in Random Walks

In a "Random Walk' with $\Delta x_t=x_t-x_{t-1}=\varepsilon_t$ , with $E(\varepsilon_t)=0$ and $E(\varepsilon_t^2)=\sigma^2$ with no autocorrelation, the coefficients of the equivalent AR (1) model would be $b_0$=0 and $b_1$=0, what will generate a mean reversion to 0/(1-0), what is equal to zero. In this way, the expected value of $\Delta x_t$ and its mean reversion level both are zero.

Doob-Meyer Submartingale Decomposition

In a submartingale $\{x_n, F_n, n = 0,1 \dots\}$ decomposed in terms of $x_n = M_n + A_n$, or the summation of a martingale M and an increasing sequence A. In this case, the increasing sequence is 'natural' for every bounded martingale, resulting in $E(M_n A_n) = E\sum_{k=1}^{n} M_{k-1}(A_k - A_{k-1})$. If martingale in unbounded, trendnents, may not be observable and increase may not be considered 'natural'.